Accelerating Variance Reduction: Optimal Complexity of Finite-Sum Optimization

Theory overview (standalone extract of the project report)

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This document reproduces the Theory section of the project report (`main.pdf`). Its results are stated with the numbers they have in the appendices of the report (the five theory notes), which contain their proofs. The bracket after each statement gives the page of the report where the appendix proves it (after "proof:"), or, for definitions, assumptions and results quoted from the literature, the page where the appendix states it. References to the other sections (Proposition 5.9 of the book and its notation are in Section 2), to the appendices and to the figures use the numbering of the full report.

4 Theory

This section states the main results of Appendices B3, B2, B1, A1 and A3, which contain the complete statements and proofs. The statements are reproduced verbatim, with the numbers they have in the appendices (equation numbers included) and without their proofs. Each ends with a bracketed pointer to the appendix: "proof:" and the page of its proof when the appendix proves it, and otherwise the page where the appendix states it (definitions, assumptions, and the results that the appendices quote from the literature without proof). In this section, as in the rest of the report, a result of the appendices is referred to by its number and its name, as in Proposition B3.16 (optimal constants of the proof structure), or by its number alone when its name or its content has just been given. Proposition 5.9, the result of the book under study, is restated in Section 2 under its number in the book.

The notation is that of Section 2, and $\mathbb{E}_{i(t)}$ is the expectation over $i(t)$ given the past. For logistic regression, $R_x = \max_i \|x_i\|$ (so $R^2 = R_x^2/4 + \lambda$, $\mu = \lambda$), $\hat{\Sigma} = \frac{1}{n}\sum_i x_i x_i^\top$, and $\ell(u) = \log(1 + e^{-u})$ is the logistic loss. Each subsection ends with what is known and what is new.

4.1 Proposition 5.9 as a chain of inequalities

Hypothesis B3 asks which of the inequalities I1–I6 of the proof of Proposition 5.9 explain the gap between its certified contraction factor $\rho_{\text{bound}} = 1 - \min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\}$ and the contraction observed along SAGA. This subsection states the instruments of the diagnosis (an exact one-step identity, a tightness ratio for each inequality, and an attribution of the gap to the inequalities) and delimits what a better analysis could certify: the best constants of the proof, a lower bound on the exponent of κ, and a trajectory-dependent version of the rate.

Exact identity and tightness ratios. Fix $t \geq 1$ and write $(\theta, z) = (\theta_{t-1}, z^{(t-1)})$ for the state before the t-th draw and $\mathcal{F}_{t-1}$ for the σ-field generated by $i(1), \dots, i(t-1)$, so that $\mathbb{E}_{i(t)} = \mathbb{E}[\cdot \mid \mathcal{F}_{t-1}]$. Let $D_{t-1} = \|\theta - \eta_*\|^2$, $M_{t-1} = \sum_i \|z_i - f_i'(\eta_*)\|^2$, $G_{t-1} = \langle F'(\theta), \theta - \eta_* \rangle$, $S_{t-1} = \sum_i \|f_i'(\theta) - f_i'(\eta_*)\|^2$, $\Omega_{t-1} = \mathbb{E}_{i(t)} \|\omega_t\|^2$ and $B_{t-1} = \mathbb{E}_{i(t)} \|b_t\|^2 = \frac{1}{n} \sum_i \|f_i'(\eta_*) - z_i + \bar{z}\|^2$, where $\bar{z}$ is the mean of the z_i and b_t is as in I1; the index $t-1$ is dropped when no confusion is possible. With a free weight $\Delta > 0$ in place of $3\gamma^2$ in (4), $V_{t-1} = D_{t-1} + \Delta M_{t-1}$. All these quantities are explicit averages over the n values of $i(t)$. Before any inequality is applied, the conditional expectation of the Lyapunov function is an exact function of them, which serves as the reference for every certified factor:

Proposition B3.5 (exact conditional contraction). *For every $t \geq 1$ and every $\gamma, \Delta > 0$, almost surely,*

$$\mathbb{E}_{i(t)}[V_t] = D_{t-1} - 2\gamma G_{t-1} + \gamma^2 \Omega_{t-1} + \Delta \Big[\Big(1 - \frac{1}{n}\Big) M_{t-1} + \frac{1}{n} S_{t-1} \Big], \tag{B3.7}$$

so that, on $\{V_{t-1} > 0\}$, the exact conditional contraction factor is

$$\rho_{\text{exact}}(t) := \frac{\mathbb{E}_{i(t)}[V_t]}{V_{t-1}} = 1 - \frac{2\gamma G_{t-1} - \gamma^2 \Omega_{t-1} + \Delta\,(M_{t-1} - S_{t-1})/n}{D_{t-1} + \Delta M_{t-1}}. \tag{B3.8}$$

[proof: Appendix B3 of the full report, p. 79]

Thus $\rho_{\text{exact}}(t)$ is computable from the state at the cost of one pass over the n components, and Section 5.3 records it along SAGA trajectories. In this notation I1–I4 read $\Omega \leq \frac{2}{n} S + 2B$, $\frac{1}{n} S \leq R^2 G$, $B \leq \frac{1}{n} M$ and $G \geq \mu D$, and I5 replaces the two coefficients left by I1–I4 by the larger one. Each inequality receives a ratio, its smaller side over its larger side, equal to 1 exactly when the inequality is an equality at the current state:

Definition B3.8 (tightness ratios). *For a state $(\theta, z) = (\theta_{t-1}, z^{(t-1)})$ with $\theta \neq \eta_*$, define, with the quantities of Section B3.2,*

$$r_{\text{sq}} := \frac{\Omega}{\frac{2}{n} S + 2B} \quad \textit{(I1)}, \qquad r_{\text{coco}} := \frac{S/n}{R^2 G} = \frac{L_{\text{eff}}}{R^2}, \quad L_{\text{eff}} := \frac{S}{nG} \quad \textit{(I2)},$$

$$r_{\text{var}} := \frac{B}{M/n} = 1 - \frac{n \|\bar{z}\|^2}{M} \quad \textit{(I3)}, \qquad r_{\text{sc}} := \frac{\mu D}{G} = \frac{\mu}{\mu_{\text{eff}}}, \quad \mu_{\text{eff}} := \frac{G}{D} \quad \textit{(I4)},$$

$$r_{\min} := \frac{\big(1 - \frac{3\mu}{16R^2}\big) D + \big(1 - \frac{1}{3n}\big) \Delta M}{\big(1 - \min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\}\big)(D + \Delta M)} \quad \textit{(I5)}, \qquad r_{\text{init}} := \frac{\sum_i \|f_i'(\theta_0) - f_i'(\eta_*)\|^2}{nR^4 \|\theta_0 - \eta_*\|^2} \quad \textit{(I6)},$$

with the convention $r_{\text{var}} := 1$ if $M = 0$ (then (I3) is the equality $0 \leq 0$; no convention is needed for r_{sq}, since $S > 0$ when $\theta \neq \eta_*$, Lemma B3.9(i)). Each ratio is the smaller side divided by the larger side of the corresponding inequality of Section B3.3 (for (I4), $G \geq \mu D$, this is $\mu D / G$); $r_{\text{sq}}, r_{\text{coco}}, r_{\text{var}}, r_{\text{sc}}$ depend on the state only (not on γ, Δ), $r_{\min}$ also depends on (Δ, n, μ, R^2).

[Appendix B3 of the full report, p. 81]

The effective constants $\mu_{\text{eff}} = G/D$ and $L_{\text{eff}} = S/(nG)$ turn I4 and I2 into identities at the current iterate. By Lemma B3.9 (ranges of the tightness ratios), for $\theta \neq \eta_*$ the ratios of I1, I2, I4 and I5 lie in $(0, 1]$, r_{var} lies in $[0, 1]$ and $\mu \leq \mu_{\text{eff}} \leq L_{\text{eff}} \leq R^2$ ($r_{\text{init}} \in (0, 1]$ when $\theta_0 \neq \eta_*$); moreover $r_{\text{sq}} = \frac{1}{2}$ exactly when the gradient changes $a_i = f_i'(\theta) - f_i'(\eta_*)$ and the memory errors $b_i = f_i'(\eta_*) - z_i + \bar{z}$ are uncorrelated under $i(t)$. As Remark B3.13 (the square-norm step discards the control variate) explains, one expects r_{sq} close to $\frac{1}{2}$ when the memory is stale and $r_{\text{sq}} \leq \frac{1}{2}$, possibly $\ll \frac{1}{2}$, when it is nearly fresh.

Representation and attribution. Inserting the ratios into (B3.7) gives the two coefficients of $\mathbb{E}_{i(t)}[V_t]$ exactly, and shows what the proof does: it substitutes worst-case values for four quantities that depend on the state.

Proposition B3.10 (exact contraction in terms of the ratios). *Let $\theta_{t-1} \neq \eta_*$ and $M_{t-1} > 0$, and let $\gamma, \Delta > 0$. Then*

$$\mathbb{E}_{i(t)}[V_t] = c_D\, D_{t-1} + c_M\, \Delta M_{t-1}, \qquad \begin{aligned} c_D &= 1 - 2\gamma\,\mu_{\text{eff}}\Big(1 - \gamma\, r_{\text{sq}} L_{\text{eff}} - \frac{L_{\text{eff}}\Delta}{2\gamma}\Big), \\ c_M &= 1 - \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\, r_{\text{sq}}\, r_{\text{var}}\Big), \end{aligned} \tag{B3.14}$$

where $\mu_{\text{eff}}, L_{\text{eff}}, r_{\text{sq}}, r_{\text{var}}$ are evaluated at the state $(\theta_{t-1}, z^{(t-1)})$. For the textbook constants $\gamma = \frac{1}{4R^2}$, $\Delta = 3\gamma^2$,

$$c_D = 1 - \frac{\mu_{\text{eff}}}{2R^2}\Big(1 - \frac{L_{\text{eff}}}{R^2}\Big(\frac{r_{\text{sq}}}{4} + \frac{3}{8}\Big)\Big), \qquad c_M = 1 - \frac{1}{n}\Big(1 - \frac{2}{3}\, r_{\text{sq}}\, r_{\text{var}}\Big). \tag{B3.15}$$

Substituting the worst-case values $r_{\text{sq}} = r_{\text{var}} = 1$, $L_{\text{eff}} = R^2$, $\mu_{\text{eff}} = \mu$ gives exactly the two coefficients $1 - \frac{3\mu}{16R^2}$ and $1 - \frac{1}{3n}$ of (B3.13). [proof: Appendix B3 of the full report, p. 82]

With the textbook constants, the parameter rate $1 - c_D$ is $\frac{\mu_{\text{eff}}}{2R^2}$ times a bracket in $[\frac{3}{8}, 1)$, so that r_{coco} and r_{sq} change it by at most a factor $\frac{8}{3}$, and the memory rate $1 - c_M$ lies between $\frac{1}{3n}$ and $\frac{1}{n}$, so that r_{sq} and r_{var} change it by at most a factor 3; only μ_{eff}/μ, which lies in $[1, \kappa]$ by Lemma B3.9 (ranges of the tightness ratios) and is therefore not bounded independently of κ, and the step I5, which replaces the weighted mean $(c_D D + c_M \Delta M)/V_{t-1}$ by $\max\{c_D, c_M\}$, can produce larger factors. Undoing the substitutions one at a time, in either of two orders, gives monotone chains from the certified to the exact factor, hence an attribution of the gap to the individual inequalities:

Proposition B3.11 (attribution of the gap). *Let $\theta_{t-1} \neq \eta_*$, $M_{t-1} > 0$, $\gamma, \Delta > 0$ with $\alpha := 1 - \gamma R^2 - \frac{R^2\Delta}{2\gamma} > 0$, and write D, M, G, S, B, Ω, V for the state quantities. Define the coefficient functions*

$$c_D(r, L, m) := 1 - 2\gamma m\Big(1 - \gamma r L - \frac{L\Delta}{2\gamma}\Big), \qquad c_M(r, v) := 1 - \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta} r v\Big),$$

and

$$\rho_{\text{bound}}(\gamma, \Delta) := \max\{c_D(1, R^2, \mu),\, c_M(1, 1)\} = 1 - \min\Big\{2\gamma\mu\alpha,\ \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\Big)\Big\},$$

which is the worst-case coefficient of Proposition B3.14 (it is < 1 iff $2\gamma^2 < \Delta$) and equals the ρ_{bound} of (I5) for the textbook constants; in the rest of this proposition we write ρ_{bound} for $\rho_{\text{bound}}(\gamma, \Delta)$.

Order A (I4 $\to$ I2 $\to$ I3 $\to$ I1 $\to$ I5; the min is kept until the end).

$$\rho_0^A := \max\{c_D(1, R^2, \mu),\, c_M(1,1)\} = \rho_{\text{bound}},$$

$$\rho_1^A := \max\{c_D(1, R^2, \mu_{\text{eff}}),\, c_M(1,1)\} = \max\Big\{1 - 2\gamma\alpha\frac{G}{D},\; 1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\Big\},$$

$$\rho_2^A := \max\{c_D(1, L_{\text{eff}}, \mu_{\text{eff}}),\, c_M(1,1)\} = \max\Big\{1 - 2\gamma\frac{G}{D} + (2\gamma^2+\Delta)\frac{S}{nD},\; 1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\Big\},$$

$$\rho_3^A := \max\{c_D(1, L_{\text{eff}}, \mu_{\text{eff}}),\, c_M(1, r_{\text{var}})\} = \max\Big\{1 - 2\gamma\frac{G}{D} + (2\gamma^2+\Delta)\frac{S}{nD},\; 1 - \frac{1}{n} + \frac{2\gamma^2 B}{\Delta M}\Big\},$$

$$\rho_4^A := \max\{c_D(r_{\text{sq}}, L_{\text{eff}}, \mu_{\text{eff}}),\, c_M(r_{\text{sq}}, r_{\text{var}})\}$$
$$= \max\Big\{1 - 2\gamma\frac{G}{D} + \frac{2\gamma^2 r_{\text{sq}} S}{nD} + \frac{\Delta S}{nD},\; 1 - \frac{1}{n} + \frac{2\gamma^2 r_{\text{sq}} B}{\Delta M}\Big\},$$

$$\rho_5^A := \frac{c_D(r_{\text{sq}}, L_{\text{eff}}, \mu_{\text{eff}})\, D + c_M(r_{\text{sq}}, r_{\text{var}})\, \Delta M}{D + \Delta M} = \rho_{\text{exact}}(t),$$

with $r_{\text{sq}} = \Omega/(2S/n + 2B)$.

Order B (I5 $\to$ I4 $\to$ I2 $\to$ I3 $\to$ I1; reverse proof order, the min is undone first).

$$\rho_0^B := \rho_{\text{bound}},$$

$$\rho_1^B := \frac{(1 - 2\gamma\mu\alpha)D + \big(1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\big)\Delta M}{V} \quad \big(= r_{\min}\,\rho_{\text{bound}} \text{ for the textbook constants}\big),$$

$$\rho_2^B := \frac{D - 2\gamma\alpha G + \big(1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\big)\Delta M}{V},$$

$$\rho_3^B := \frac{D - 2\gamma G + (2\gamma^2+\Delta)\frac{S}{n} + \frac{2\gamma^2}{n}M + \big(1 - \frac{1}{n}\big)\Delta M}{V},$$

$$\rho_4^B := \frac{D - 2\gamma G + (2\gamma^2+\Delta)\frac{S}{n} + 2\gamma^2 B + \big(1 - \frac{1}{n}\big)\Delta M}{V},$$

$$\rho_5^B := \frac{D - 2\gamma G + \gamma^2\Omega + \Delta\frac{S}{n} + \big(1 - \frac{1}{n}\big)\Delta M}{V} = \rho_{\text{exact}}(t).$$

Then both sequences are nonincreasing:

$$\rho_{\text{bound}} = \rho_0^A \ge \rho_1^A \ge \rho_2^A \ge \rho_3^A \ge \rho_4^A \ge \rho_5^A = \rho_{\text{exact}}(t), \qquad \rho_{\text{bound}} = \rho_0^B \ge \rho_1^B \ge \cdots \ge \rho_5^B = \rho_{\text{exact}}(t),$$

and consequently the nonnegative increments $\delta_k := \rho_{k-1} - \rho_k$ *(additive attribution) and, when* $\rho_{\text{exact}}(t) > 0$, $\nu_k := \log(\rho_{k-1}/\rho_k)$ *(attribution in nats per iteration) satisfy*

$$\sum_{k=1}^{5}\delta_k = \rho_{\text{bound}} - \rho_{\text{exact}}(t), \qquad \sum_{k=1}^{5}\nu_k = \log\frac{1}{\rho_{\text{exact}}(t)} - \log\frac{1}{\rho_{\text{bound}}}, \qquad \prod_{k=1}^{5}\frac{\rho_k}{\rho_{k-1}} = \frac{\rho_{\text{exact}}(t)}{\rho_{\text{bound}}}.$$

In order A the increments are attributed to (I4), (I2), (I3), (I1), (I5) in this order; in order B to (I5), (I4), (I2), (I3), (I1). [proof: Appendix B3 of the full report, p. 83]

The increments ν_k add up to the gap between the exact and the certified contraction exponents, and the *share* of an inequality is its increment divided by this gap. The shares depend on the order. Order B gives each inequality its full effect on the weighted contraction; in order A, undoing I4 and I2, which only lower c_D, has no effect as long as c_M is the larger coefficient ($\kappa < \frac{9n}{16}$ for the textbook constants), and undoing I3, which only lowers c_M, has none as long as c_D is. By Remark B3.12 (reading the two attribution orders), comparing the two orders shows which component of V_t is the bottleneck of the bound. Section 5.3 measures the ratios and the shares along SAGA trajectories on logistic regression.

Optimal constants and the exponent of κ. The identities and I1–I3 do not involve γ or Δ, so the same chain certifies a rate for every pair (γ, Δ) of an explicit admissible set; optimizing over this set measures how much of the gap is due to the constants $\gamma = \frac{1}{4R^2}$, $\Delta = 3\gamma^2$ of Proposition 5.9.

Proposition B3.14 (certified rate for general (γ, Δ))**.** *Let $\gamma, \Delta > 0$ satisfy*

$$\alpha(\gamma, \Delta) := 1 - \gamma R^2 - \frac{R^2 \Delta}{2\gamma} > 0 \qquad \text{and} \qquad \frac{2\gamma^2}{\Delta} < 1.$$

Then for all $t \geq 1$, almost surely,

$$\mathbb{E}_{i(t)}[V_t] \leq \Big(1 - \min\Big\{2\gamma\mu\,\alpha(\gamma, \Delta),\ \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\Big)\Big\}\Big) V_{t-1}, \tag{B3.16}$$

and both arguments of the min lie in $(0, \frac{1}{2}]$ and $(0, \frac{1}{n})$ respectively.

[proof: Appendix B3 of the full report, p. 85]

Writing $\Delta = c\gamma^2$ separates the two constants: the memory rate depends on c only, and for fixed c the parameter rate is a concave function of the step.

Corollary B3.15 (parametrization $\Delta = c\gamma^2$)**.** *Write $\Delta = c\gamma^2$ with $c > 2$. Then $\alpha = 1 - \gamma R^2(1 + \frac{c}{2})$, the condition $\alpha > 0$ reads $\gamma < \frac{2}{R^2(2+c)}$, and (B3.16) becomes*

$$\begin{aligned} &\mathbb{E}_{i(t)}[V_t] \leq \big(1 - \min\{\tau_D(\gamma, c), \tau_M(c)\}\big) V_{t-1}, \\ &\tau_D(\gamma, c) := 2\gamma\mu\Big(1 - \gamma R^2\Big(1 + \frac{c}{2}\Big)\Big), \qquad \tau_M(c) := \frac{1 - 2/c}{n}. \end{aligned} \tag{B3.17}$$

For fixed c, $\gamma \mapsto \tau_D(\gamma, c)$ is a concave parabola, maximal at

$$\gamma^\star(c) = \frac{1}{R^2(2+c)}, \qquad \tau_D(\gamma^\star(c), c) = \frac{\mu}{R^2(2+c)} = \frac{1}{\kappa(2+c)},$$

while τ_M does not depend on γ. Hence the best rate that (B3.17) certifies for a given c is

$$\tau(c) := \min\Big\{\frac{1}{\kappa(2+c)},\ \frac{c-2}{cn}\Big\}.$$

[proof: Appendix B3 of the full report, p. 86]

Since $c > 2$, no weight is admissible when $\gamma R^2 \geq \frac{1}{2}$. At the textbook step every $c \in (2, 6)$ is admissible, and the best of these weights certifies a rate asymptotically equivalent to $\frac{2}{3n}$, twice the textbook $\frac{1}{3n}$, as $n/\kappa \to \infty$ (Section B3.5). Maximizing $\tau(c)$ over c gives the best constants of the proof structure in closed form:

Proposition B3.16 (optimal constants of the proof structure)**.** *Let $s := n/\kappa = n\mu/R^2$. The maximum over $c > 2$ and admissible γ of the rate $\min\{\tau_D, \tau_M\}$ in (B3.17) is attained at*

$$\begin{aligned} c^\star &= \frac{s + \sqrt{s^2 + 16}}{2}, \qquad \Delta^\star = c^\star(\gamma^\star)^2, \\ \gamma^\star &= \frac{1}{R^2(2 + c^\star)} = \frac{2}{R^2\big(4 + s + \sqrt{s^2 + 16}\big)} = \frac{2}{4R^2 + n\mu + \sqrt{n^2\mu^2 + 16R^4}}, \end{aligned}$$

and equals

$$\tau^\star = \frac{1}{\kappa(2 + c^\star)} = \frac{2}{4\kappa + n + \sqrt{n^2 + 16\kappa^2}}, \qquad \frac{1}{n + 4\kappa} \ \leq\ \tau^\star \ \leq\ \frac{4 - 2\sqrt{2}}{n + 4\kappa} \approx \frac{1.172}{n + 4\kappa}.$$

In particular $\tau^\star \sim \frac{1}{4\kappa}$, $\gamma^\star \sim \frac{1}{4R^2}$ and $c^\star \to 2$ when $n/\kappa \to 0$, and $\tau^\star \sim \frac{1}{n}$, $\gamma^\star \sim \frac{1}{n\mu}$ and $c^\star \sim n/\kappa$ when $n/\kappa \to \infty$.

[proof: Appendix B3 of the full report, p. 86]

These are the optimal rate and step of [20, Theorem 1 and Corollary 1] for SAGA: their Lyapunov function has the same two-term form, but its memory term averages upper bounds $H_i \geq \|z_i - f_i'(\eta_*)\|^2$ obtained from Bregman divergences, and their proof uses the same I1 and I3 but applies strong convexity and cocoercivity through function values; the two chains differ, but their optima coincide (Section B3.5). The ratio of $\tau^\star$ to the textbook rate $\min\{\frac{1}{3n}, \frac{3}{16\kappa}\}$ lies in $[1.05, 3)$. For $\kappa \geq \frac{9n}{16}$ it is at most $\frac{4}{3}$: there the textbook constants are within $\frac{4}{3}$ of the best of the proof structure (the ratio is $\frac{32}{3(5+\sqrt{17})} \approx 1.17$ for $n = 1000$, $\kappa = 10^3$; Section 5.4). For $n = 1000$ and $\kappa \leq 10^2$ the constants matter more: the optimal constants, at a smaller step, raise the certified rate by a factor 2.42 ($\kappa = 10^2$) or 2.94 ($\kappa = 10$), of the same size as the speed-ups 3.2–3.7 of the exact over the certified exponent measured there at the textbook step (Section 5.3), as noted in Remark B3.18 (optimizing (γ, Δ) gains at most a factor 3). At the textbook step itself, the best weight Δ gains a factor 1.81 or 1.98 (Section 5.4). This optimality is relative to the proof structure (Lyapunov function $D + \Delta M$, inequalities I1–I5): another Lyapunov function or another chain could certify more. Whatever the analysis, however, this Lyapunov function cannot certify more than $\frac{1}{n}$ per iteration: the memory recursion (B3.5) of Lemma B3.4 (exact one-step identities), $\mathbb{E}_{i(t)}[M_t] = (1 - \frac{1}{n})M_{t-1} + \frac{1}{n}S_{t-1}$, gives $\mathbb{E}_{i(t)}[M_t] \geq (1 - \frac{1}{n})M_{t-1}$, hence $\mathbb{E}[V_t] \geq \Delta(1 - \frac{1}{n})^t M_0$, and no bound $\mathbb{E}[V_t] \leq c_0 \rho^t V_0$ valid for all t, with c_0 independent of t and $M_0 > 0$, has $1 - \rho > \frac{1}{n}$, whatever $\Delta > 0$. When $\kappa \leq \frac{9n}{16}$, where the textbook rate is $\frac{1}{3n}$, a factor 3 is thus the most that any analysis of $\mathbb{E}[V_t]$ can gain.

Conversely, the exponent 1 of κ is intrinsic to SAGA with a constant step, not to its analysis. The following instance satisfies Assumption B3.1 (convex smooth components, strongly convex sum), the setting of Section 2, and (B3.1) in its statement is the SAGA iteration (3):

Proposition B3.17 (lower bound for SAGA with any constant step)**.** *Let $n \geq 1$, $0 < \mu < R^2$, $d = 2$, and $f_i(\theta) = \frac{1}{2}(\mu\theta_1^2 + R^2\theta_2^2)$ for all $i \leq n$. Assumption B3.1 holds with these constants, $F = f_1$, $F'(\theta) = Q\theta$ with $Q = \operatorname{diag}(\mu, R^2)$, and $\eta_* = 0$. Run SAGA (B3.1) with any constant step $\gamma > 0$ (which may depend on μ) and any deterministic initial memory $z^{(0)}$. Then for all $t \geq 0$,*

$$\mathbb{E}[\theta_t] = (I - \gamma Q)^t \theta_0 \qquad \text{and} \qquad \mathbb{E}\,\|\theta_t - \eta_*\|^2 \;\geq\; \|\mathbb{E}[\theta_t] - \eta_*\|^2 \;\geq\; \Big(\frac{\kappa - 1}{\kappa + 1}\Big)^{2t} \min\{\theta_{0,1}^2, \theta_{0,2}^2\}. \tag{B3.18}$$

In particular, for $\theta_0 = (r, r)$ with $r \neq 0$ and $\varepsilon < \frac{1}{2}$, reaching $\mathbb{E}\,\|\theta_t - \eta_*\|^2 \leq \varepsilon\,\|\theta_0 - \eta_*\|^2$ requires $t \geq \frac{\kappa - 1}{4} \log \frac{1}{2\varepsilon}$, whatever the step. [proof: Appendix B3 of the full report, p. 88]

The argument only uses that the direction is unbiased and that F' is affine: on any quadratic F the mean SAGA iterate is the gradient-descent iterate with the same step, whatever the memory, and the variance only adds to the error. For $\kappa \gg n$ the textbook bound is within the factor $\frac{64}{3}$ of this bound on its leading term, and within $\frac{8}{3}$ of the bound $(2\kappa - \frac{1}{2}) \log \frac{1}{2\varepsilon}$ that the same argument gives for the textbook step; caveat (ii) of Remark B3.18 derives both comparisons.

Trajectory-dependent version. Strong convexity I4 and cocoercivity I2 are only used at the current iterate, so the proof also certifies rates in which effective constants replace μ and R^2; this is the hook for hypothesis B2 (Section 4.2). Regard $\mu_{\text{eff}}(\theta) = \langle F'(\theta), \theta - \eta_* \rangle \,/\, \|\theta - \eta_*\|^2$ and $L_{\text{eff}}(\theta) = \frac{1}{n}\sum_i \|f_i'(\theta) - f_i'(\eta_*)\|^2 \,/\, \langle F'(\theta), \theta - \eta_* \rangle$ as functions of $\theta \neq \eta_*$ (they do not depend on the memory), and write $\mu_{\text{eff}}(t-1) = \mu_{\text{eff}}(\theta_{t-1})$ and $L_{\text{eff}}(t-1) = L_{\text{eff}}(\theta_{t-1})$:

Proposition B3.19 (a-posteriori rates with effective constants)**.** *Let $\gamma = \frac{1}{4R^2}$ and $\Delta = 3\gamma^2$. For*

every $t \geq 1$, almost surely on $\{\theta_{t-1} \neq \eta_*\}$:

$$\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \leq \Big(1 - \min\Big\{\frac{1}{3n},\ \frac{3\,\mu_{\text{eff}}(t-1)}{16R^2}\Big\}\Big)V_{t-1}, \tag{B3.19}$$

$$\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \leq \Big(1 - \min\Big\{\frac{1}{3n},\ \frac{\mu_{\text{eff}}(t-1)}{2R^2}\Big(1 - \frac{5\,L_{\text{eff}}(t-1)}{8R^2}\Big)\Big\}\Big)V_{t-1}, \tag{B3.20}$$

and the coefficient in (B3.20) satisfies $\frac{3\mu_{\text{eff}}(t-1)}{16R^2} \leq \frac{\mu_{\text{eff}}(t-1)}{2R^2}\big(1 - \frac{5L_{\text{eff}}(t-1)}{8R^2}\big) \leq \frac{\mu_{\text{eff}}(t-1)}{2R^2} \leq \frac{1}{2}$, so that both contraction factors lie in $[\frac{1}{2}, 1)$. More generally, for γ, Δ as in Proposition B3.14,

$$\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \leq \Big(1 - \min\Big\{2\gamma\,\mu_{\text{eff}}(t-1)\Big(1 - \gamma L_{\text{eff}}(t-1) - \frac{L_{\text{eff}}(t-1)\Delta}{2\gamma}\Big),\ \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\Big)\Big\}\Big)V_{t-1}. \tag{B3.21}$$

On $\{\theta_{t-1} = \eta_*\}$ one has $\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \leq \big(1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\big)V_{t-1}$, so that (B3.19)–(B3.21) hold there with any value assigned to $\mu_{\text{eff}}(t-1)$ and $L_{\text{eff}}(t-1)$. [proof: Appendix B3 of the full report, p. 90]

In the second bound, which uses both effective constants, μ_{eff} enters linearly, whereas a small L_{eff} helps by at most a factor $\frac{8}{3}$, as explained in Remark B3.20 (the step is fixed in advance). These factors are random ($\mathcal{F}_{t-1}$-measurable) and correlated with V_{t-1}, so the conditional bounds cannot be iterated: the product of the factors along a trajectory, which Figure 1 plots, does not bound $\mathbb{E}[V_T]$ in general. An a-priori rate needs a deterministic bound on the effective constants over a region Θ of the parameter space, and follows from the supermartingale $\rho_\Theta^{-(t\wedge\tau)}V_{t\wedge\tau}$, where τ now denotes the exit time from Θ (not a rate as in $\tau(c)$ and $\tau^\star$):

Proposition B3.21 (a-priori rate on a region). *Let $\Theta \subset \mathbb{R}^d$ and*

$$\mu_\Theta := \inf_{\theta\in\Theta\setminus\{\eta_*\}} \mu_{\text{eff}}(\theta) \ \geq \mu, \qquad L_\Theta := \sup_{\theta\in\Theta\setminus\{\eta_*\}} L_{\text{eff}}(\theta) \ \leq R^2,$$

$$\rho_\Theta := 1 - \min\Big\{\frac{1}{3n},\ \frac{\mu_\Theta}{2R^2}\Big(1 - \frac{5L_\Theta}{8R^2}\Big)\Big\} \ \leq \ 1 - \min\Big\{\frac{1}{3n}, \frac{3\mu_\Theta}{16R^2}\Big\},$$

and let $\tau := \inf\{t \geq 0 : \theta_t \notin \Theta\}$ (with $\inf\emptyset = +\infty$). For $\gamma = \frac{1}{4R^2}$, $\Delta = 3\gamma^2$ and every $T \geq 0$,

$$\mathbb{E}\big[V_T\,\mathbf{1}_{\{\tau>T\}}\big] \leq \rho_\Theta^T\,V_0. \tag{B3.22}$$

In particular, if $\theta_t \in \Theta$ for all t almost surely, then $\mathbb{E}[V_T] \leq \rho_\Theta^T V_0$ and $\mathbb{E}\,\|\theta_T - \eta_*\|^2 \leq \rho_\Theta^T(1 + \frac{n}{4})\,\|\theta_0 - \eta_*\|^2$ for the book's initialization. [proof: Appendix B3 of the full report, p. 91]

For logistic regression, the self-concordance-type bound $|\ell'''| \leq \ell''$ [6, 8] gives an explicit lower bound on μ_Θ over balls around η_*:

Proposition B3.22 (local strong convexity of logistic regression). *Let $f_i(\theta) = \ell(y_i x_i^\top\theta) + \frac{\lambda}{2}\,\|\theta\|^2$ as in Remark B3.7, $\lambda > 0$, $R_x := \max_i \|x_i\|$, and $\psi(x) := \frac{1-e^{-x}}{x}$ for $x > 0$, $\psi(0) := 1$ (ψ is decreasing on $[0,\infty)$, with values in $(0,1]$). Then for every $\theta \neq \eta_*$,*

$$\mu_{\text{eff}}(\theta) \ \geq \ \lambda + \psi\big(R_x\,\|\theta - \eta_*\|\,\big)\big(\lambda_{\min}(F''(\eta_*)) - \lambda\big) \ \geq \ \max\big\{\lambda,\ \psi\big(R_x\,\|\theta - \eta_*\|\,\big)\lambda_{\min}(F''(\eta_*))\big\},$$

where $F''(\eta_*) = \lambda I + \frac{1}{n}\sum_i \ell''(y_i x_i^\top\eta_*)x_i x_i^\top \succeq \lambda I$. Consequently, on the ball $\Theta_r = \{\theta : \|\theta - \eta_*\| \leq r\}$, the constant μ_{Θ_r} of Proposition B3.21 satisfies

$$\mu_{\Theta_r} \geq \lambda + \psi(R_x r)(\lambda_{\min}(F''(\eta_*)) - \lambda) \geq \max\{\lambda, \psi(R_x r)\lambda_{\min}(F''(\eta_*))\};$$

in particular $\mu_{\Theta_r} \geq \frac{1}{2}(\lambda + \lambda_{\min}(F''(\eta_*))) \geq \max\{\lambda, \frac{1}{2}\lambda_{\min}(F''(\eta_*))\}$ as soon as $R_x r \leq \frac{3}{2}$ (since $\psi(3/2) \approx 0.518$). [proof: Appendix B3 of the full report, p. 92]

Applied from the first time the iterate enters the ball of radius $\frac{3}{2R_x}$ around η_* (by the strong Markov property of $(\theta_t, z^{(t)})$, with the Lyapunov value at that time in place of V_0), Proposition B3.21 (a-priori rate on a region) gives, until the next exit, a contraction factor at most $1-\min\{\frac{1}{3n}, \frac{3}{32R^2}(\lambda+\lambda_{\min}(F''(\eta_*)))\}$: the parameter term of Proposition 5.9 is multiplied by $\frac{1}{2}(1+\lambda_{\min}(F''(\eta_*))/\lambda)$, which changes the rate only when that term binds ($\kappa > \frac{9n}{16}$). From $\theta_0 = 0$, the starting point lies outside this ball in all our problems with $\kappa \geq 10^2$ ($\|\eta_*\| \geq 5.9$, $R_x = 1$). In the fifteen configurations of Section 5.3, $\lambda_{\min}(F''(\eta_*))/\lambda$ ranges from 1.0003 to 69.9 and is below 2 in nine of them: six have $\kappa \leq 10^2$, where the memory term binds and the rate is unchanged, and the other three gain less than a factor $\frac{3}{2}$.

Known and new. Proposition 5.9 and its proof are those of [9]; Lyapunov functions adding memory errors, or upper bounds on them, to the squared distance are used in [20, 21]; the optimal constants of this proof structure (Proposition B3.16) are those of [20]; for oblivious step schedules depending only on R^2, the exponent 1 of κ follows from [5, Theorem 2]. To our knowledge, the use of the exact identity as an instrument, the reading of the proof as a substitution of worst-case values with monotone attribution chains, the elementary lower bound for every n and every constant step, and the trajectory-dependent bounds are not in the sources we reviewed in this form.

4.2 Effective curvature and smoothness along trajectories

Hypothesis B2 asks whether the constants of the strong-convexity step I4 and of the cocoercivity step I2, evaluated along the trajectory actually followed by the algorithm, describe its convergence better than their worst-case values μ and R^2. Appendix B2 works under the following assumption, in which L is the smoothness constant L_F of Section 2, and defines the two constants that make I4 and I2 identities at the current iterate.

Assumption B2.1 (convex R^2-smooth f_i, μ-strongly convex F). *(i) Each $f_i : \mathbb{R}^d \to \mathbb{R}$ is convex, differentiable and R^2-smooth, i.e. f_i' is R^2-Lipschitz-continuous.*

(ii) $F = \frac{1}{n}\sum_{i=1}^n f_i$ is μ-strongly convex with $\mu > 0$; its (unique) minimizer is η_*, so that $F'(\eta_*) = 0$ (Proposition 5.2 of [9]). We denote by $L \leq R^2$ the smoothness constant of F (F' is L-Lipschitz-continuous; $L \leq R^2$ because an average of R^2-Lipschitz maps is R^2-Lipschitz), and by $\kappa = R^2/\mu$ the condition number used in the book.

(iii) Whenever second derivatives appear (Lemma B2.5, Proposition B2.8, Section B2.3 and after), each f_i is twice continuously differentiable.

[Appendix B2 of the full report, p. 95]

Definition B2.2 (effective constants μ_{eff} and L_{eff}). *For $\theta \neq \eta_*$, define*

$$\mu_{\text{eff}}(\theta) \;=\; \frac{(\theta-\eta_*)^\top F'(\theta)}{\|\theta-\eta_*\|^2}, \tag{B2.1}$$

$$L_{\text{eff}}(\theta) \;=\; \frac{\frac{1}{n}\sum_{i=1}^n \|f_i'(\theta)-f_i'(\eta_*)\|^2}{(\theta-\eta_*)^\top F'(\theta)}. \tag{B2.2}$$

Along a trajectory $(\theta_t)_{t\geq 0}$ we abbreviate $\mu_{\text{eff}}(t) = \mu_{\text{eff}}(\theta_t)$ and $L_{\text{eff}}(t) = L_{\text{eff}}(\theta_t)$.

[Appendix B2 of the full report, p. 95]

Both quotients are well defined and positive, by Lemma B2.4 (μ_{eff} lies between μ and L) and by Lemma B2.6 below, and μ/μ_{eff} and L_{eff}/R^2 are the tightness ratios r_{sc} and r_{coco} of I4 and I2 in Definition B3.8 (tightness ratios).

Ranges and directional limits. For C^2 functions, μ_{eff} is the curvature, in the direction of the error, of an averaged Hessian, and both constants lie between the worst-case ones.

Lemma B2.5 (integral form of the effective curvature). *Assume F is twice continuously differentiable and let, for $\theta \neq \eta_*$,*

$$\bar{H}(\theta) \;=\; \int_0^1 F''(\eta_* + s\,v)\,ds \qquad (v = \theta - \eta_*,\; u = v/\|v\|).$$

Then $\bar{H}(\theta)$ is symmetric, $\mu I \preccurlyeq \bar{H}(\theta) \preccurlyeq LI$, $F'(\theta) = \bar{H}(\theta)\,v$, and

$$\mu_{\text{eff}}(\theta) \;=\; \frac{v^\top \bar{H}(\theta)\,v}{\|v\|^2} \;=\; u^\top \bar{H}(\theta)\,u, \tag{B2.5}$$

i.e. $\mu_{\text{eff}}(\theta)$ is the Rayleigh quotient, in the direction of the error, of the Hessian averaged along the segment $[\eta_*, \theta]$. [proof: Appendix B2 of the full report, p. 97]

Lemma B2.6 (L_{eff} lies between μ_{eff} and R^2). *Under Assumption B2.1, for all $\theta \neq \eta_*$, with $v = \theta - \eta_*$,*

$$\|F'(\theta)\|^2 \;\le\; \frac{1}{n}\sum_{i=1}^n \|f_i'(\theta) - f_i'(\eta_*)\|^2 \qquad \textit{(Jensen / gradient diversity)}, \tag{B2.6}$$

$$\frac{1}{n}\sum_{i=1}^n \|f_i'(\theta) - f_i'(\eta_*)\|^2 \;\le\; R^2\, v^\top F'(\theta) \qquad \textit{(averaged co-coercivity, eq. (5.31) of the book)}, \tag{B2.7}$$

$$\mu \;\le\; \mu_{\text{eff}}(\theta) \;\le\; \frac{\|F'(\theta)\|}{\|v\|} \;\le\; \frac{\|F'(\theta)\|^2}{v^\top F'(\theta)}. \tag{B2.8}$$

Consequently

$$\mu \;\le\; \mu_{\text{eff}}(\theta) \;\le\; \frac{\|F'(\theta)\|^2}{v^\top F'(\theta)} \;\le\; L_{\text{eff}}(\theta) \;\le\; R^2, \tag{B2.9}$$

and, more precisely, with $h_i = f_i'(\theta) - f_i'(\eta_*)$ (the vectors a_i of Section B3.2) and $\bar{h} = \frac{1}{n}\sum_i h_i = F'(\theta)$,

$$L_{\text{eff}}(\theta) \;=\; \frac{\|F'(\theta)\|^2 + \frac{1}{n}\sum_{i=1}^n \|h_i - \bar{h}\|^2}{v^\top F'(\theta)}. \tag{B2.10}$$

If moreover each f_i is L_i-smooth, then $L_{\text{eff}}(\theta) \le \max_i L_i$ (so one may take $R^2 = \max_i L_i$).
[proof: Appendix B2 of the full report, p. 97]

The effective condition number $L_{\text{eff}}/\mu_{\text{eff}}$ is thus at least 1, and (B2.10) writes L_{eff}, and hence the tightness ratio $r_{\text{coco}} = L_{\text{eff}}/R^2$ of I2, as the sum of two terms: the diversity of the gradient differences h_i, which vanishes iff all h_i coincide, and the part of R^2 that the full gradient uses in the direction of the error. Near the optimum, the two constants become functions of this direction.

Proposition B2.8 (directional limits at the optimum). *Assume each f_i is twice continuously differentiable, and write $H_* = F''(\eta_*)$ and $M_* = \frac{1}{n}\sum_{i=1}^n f_i''(\eta_*)^2$. For a unit vector u define*

$$\Lambda(u) \;=\; \frac{u^\top M_* u}{u^\top H_* u}.$$

Then, as $\theta \to \eta_*$ (with $u = (\theta - \eta_*)/\|\theta - \eta_*\|$ allowed to vary),

$$\mu_{\text{eff}}(\theta) - u^\top H_* u \;\to\; 0, \qquad L_{\text{eff}}(\theta) - \Lambda(u) \;\to\; 0,$$

uniformly in the direction. In particular, if $\theta \to \eta_*$ along a fixed direction u, then $\mu_{\text{eff}}(\theta) \to u^\top H_* u$ (the Rayleigh quotient of the Hessian at the optimum in the direction of approach) and $L_{\text{eff}}(\theta) \to \Lambda(u)$. Moreover, for every unit u,

$$u^\top H_* u \;\le\; \frac{u^\top H_*^2 u}{u^\top H_* u} \;\le\; \Lambda(u) \;\le\; R^2, \tag{B2.11}$$

consistently with (B2.9). [proof: Appendix B2 of the full report, p. 98]

As u ranges over the unit sphere, $u^\top H_* u$ spans $[\lambda_{\min}(H_*), \lambda_{\max}(H_*)]$ and equals the local strong-convexity constant $\lambda_{\min}(H_*)$ only on the bottom eigenspace of H_*: near η_*, the slack of I4 is set by the direction in which the algorithm approaches η_*.

Logistic regression. For ℓ_2-regularized logistic regression, the global strong-convexity constant μ_{glob} (the largest μ for which F is μ-strongly convex) is exactly λ, whatever the data, but, when the features span $\mathbb{R}^d$, the value λ is approached only at infinity.

Proposition B2.12 (the global strong convexity constant is λ)**.** *For ℓ_2-regularized logistic regression, let $I_0 = \{i : x_i \neq 0\}$ (if $I_0 = \emptyset$ then $F'' \equiv \lambda I$ and there is nothing to prove).*

(a) There exists a unit vector $\hat{d} \in \mathbb{R}^d$ with $x_i^\top \hat{d} \neq 0$ for every $i \in I_0$; in fact Lebesgue-almost every direction has this property.

(b) For every such $\hat{d}$, with $\delta(\hat{d}) = \min_{i \in I_0} \left|x_i^\top \hat{d}\right| > 0$, and every $t \ge 0$,

$$\lambda \;\le\; \lambda_{\min}\big(F''(t\hat{d})\big) \;\le\; \lambda + R_x^2\, e^{-t\,\delta(\hat{d})}. \tag{B2.20}$$

(c) Consequently $\mu_{\text{glob}} = \inf_\theta \lambda_{\min}(F''(\theta)) = \lambda$: the global strong convexity constant of F is exactly λ, whatever the data.

(d) The infimum is attained at some θ if and only if $\operatorname{span}\{x_1, \dots, x_n\} \neq \mathbb{R}^d$; in that case $\lambda_{\min}(F''(\theta)) = \lambda$ for every θ. If the features span $\mathbb{R}^d$, then $\lambda_{\min}(F''(\theta)) > \lambda$ for every θ and the value λ is only approached as $\|\theta\| \to \infty$; it is in fact approached along every *ray: for every unit vector $\hat{d}$ and every $t \ge 0$,*

$$\lambda_{\min}\big(F''(t\hat{d})\big) \;\le\; \lambda + \frac{1}{n}\sum_{i=1}^{n} \ell''\big(t\, y_i x_i^\top \hat{d}\big)(x_i^\top \hat{d})^2 \;\le\; \lambda + R_x^2\, e^{-t\,\delta'(\hat{d})}, \tag{B2.21}$$

where $\delta'(\hat{d}) = \min\{\left|x_i^\top \hat{d}\right| : x_i^\top \hat{d} \neq 0\}$, with the conventions $\delta'(\hat{d}) = +\infty$ and $e^{-\infty} = 0$ when $x_i^\top \hat{d} = 0$ for all i; hence $\lim_{t\to\infty} \lambda_{\min}(F''(t\hat{d})) = \lambda$ for every direction $\hat{d}$. Directions as in (a) are those along which the whole matrix $F''(t\hat{d}) - \lambda I$ tends to 0; along a general ray,

$$F''(t\hat{d}) - \lambda I \;\xrightarrow[t\to\infty]{}\; \frac{1}{4n} \sum_{i:\, x_i^\top \hat{d} = 0} x_i x_i^\top,$$

a matrix which is singular because $\hat{d}$ lies in its kernel.

[proof: Appendix B2 of the full report, p. 101]

Corollary B2.13 (strong convexity on a bounded region). *In ℓ_2-regularized logistic regression, for every $D > 0$ and every θ with $\|\theta\| \le D$,*

$$\lambda_{\min}\big(F''(\theta)\big) \;\ge\; \lambda + \ell''(R_x D)\,\lambda_{\min}(\hat{\Sigma}) \;\ge\; \lambda + \tfrac{1}{4}e^{-R_x D}\lambda_{\min}(\hat{\Sigma}). \tag{B2.22}$$

Moreover $\|\eta_*\| \le \sqrt{2\log 2/\lambda}$, so that the ball $\{\|\theta\| \le 2\sqrt{2\log 2/\lambda}\}$ contains the ball of radius $\sqrt{2\log 2/\lambda}$ around η_*. [proof: Appendix B2 of the full report, p. 102]

An algorithm whose iterates stay in a bounded region thus never experiences the curvature λ unless $\lambda_{\min}(\hat{\Sigma}) = 0$. Gradient descent with $\gamma \le 1/L$ stays in the ball of radius $\|\theta_0 - \eta_*\|$ around η_*; with the step and initialization of Proposition 5.9, the SAGA iterates are almost surely bounded but not, in general, confined to a deterministic ball (Section B2.3). Near η_*, Lemma B2.14 (generalized self-concordance of the logistic loss) yields a comparison of μ_{eff} with the curvature at the optimum in the direction of the error.

Corollary B2.15 (effective curvature versus curvature at the optimum). *For ℓ_2-regularized logistic regression, let $p_i = \sigma'(y_i x_i^\top \eta_*)$, $H_* = F''(\eta_*) = \frac{1}{n}\sum_i p_i x_i x_i^\top + \lambda I$ and $p_{\min} = \min_i p_i$. For every $\theta \neq \eta_*$, with $u = (\theta - \eta_*)/\|\theta - \eta_*\|$,*

$$\lambda + e^{-R_x\|\theta-\eta_*\|}\left(u^\top H_* u - \lambda\right) \;\le\; \mu_{\text{eff}}(\theta), \tag{B2.24}$$

$$\mu_{\text{eff}}(\theta) \;\le\; \lambda + e^{R_x\|\theta-\eta_*\|}\left(u^\top H_* u - \lambda\right), \tag{B2.25}$$

and

$$u^\top H_* u \;\ge\; \lambda + p_{\min}\, u^\top \hat{\Sigma} u \;\ge\; \lambda + p_{\min}\lambda_{\min}(\hat{\Sigma}). \tag{B2.26}$$

In particular, $\liminf_{\theta\to\eta_*} \mu_{\text{eff}}(\theta) \ge \lambda + p_{\min}\lambda_{\min}(\hat{\Sigma})$, and for $\|\theta - \eta_*\| \le 1/R_x$ one has $\mu_{\text{eff}}(\theta) \ge \lambda + e^{-1}p_{\min}\lambda_{\min}(\hat{\Sigma})$. If moreover $\theta \in S = \operatorname{span}\{x_1, \dots, x_n\}$ (which holds along the whole trajectory of gradient descent and of SAGA when $\theta_0 \in S$, Lemma B2.16), then $u \in S$, and $\lambda_{\min}(\hat{\Sigma})$ may be replaced, in (B2.26) and in the two consequences, by $\lambda^+_{\min}(\hat{\Sigma})$, the smallest nonzero eigenvalue of $\hat{\Sigma}$ (which is positive even when $d > n$). [proof: Appendix B2 of the full report, p. 103]

Once $R_x\|\theta_t - \eta_*\| \le 1$, $\mu_{\text{eff}}(\theta_t)/\lambda$ is therefore within a factor $e^{\pm 1}$ of $u_t^\top H_* u_t/\lambda$, where u_t is the direction of the error. This ratio is close to 1 in every direction under strong regularization, $\lambda \gg \lambda_{\max}(\hat{\Sigma})/4$, which forces $\kappa \ll n$; under intermediate regularization, $\lambda_{\min}(H_* - \lambda I) \ll \lambda \ll \lambda_{\max}(H_* - \lambda I)$, typical of anisotropic features, it is close to 1 along the bottom eigenvectors of H_* and of order $\lambda_{\max}(H_*)/\lambda$ along the top ones; under weak regularization with well-conditioned features, $\lambda \ll p_{\min}\lambda_{\min}(\hat{\Sigma})$, it is large in every direction (Section B2.3).

Which curvature does an algorithm meet? On a quadratic, gradient descent concentrates its error on the bottom eigenspace of the Hessian, unless the initial error is orthogonal to it. In the statement, $\pi_k(t) = (q_k^\top v_t)^2/\|v_t\|^2$ is the share of $\|v_t\|^2$ carried by the eigenvector q_k (the spectral profile (B2.28)).

Proposition B2.17 (alignment of gradient descent with the flattest direction). *Let $F(\theta) = F(\eta_*) + \frac{1}{2}(\theta-\eta_*)^\top H(\theta-\eta_*)$, where H is symmetric positive definite with eigenvalues $\lambda_1 \ge \cdots \ge \lambda_d = \lambda_{\min} > 0$ and orthonormal eigenvectors $q_1, \dots, q_d$; let $E_{\min}$ be the eigenspace of $\lambda_{\min}$, P the orthogonal projector onto $E_{\min}$, and $\lambda' = \min\{\lambda_k : \lambda_k > \lambda_{\min}\}$ the second smallest distinct eigenvalue (if $H = \lambda_{\min} I$, then $P = I$ and the statement is trivial). Let $\theta_t = \theta_{t-1} - \gamma F'(\theta_{t-1})$ be gradient descent,*

$v_t = \theta_t - \eta_*$, *and* $\pi_k(t)$ *the spectral profile* (B2.28) *of the error. Let* $0 < \gamma \le 1/\lambda_1$ *and* $v_0 \neq 0$*; if* $\gamma = 1/\lambda_1$*, assume moreover that* v_0 *does not lie in the eigenspace of* λ_1 *(otherwise* $\theta_t = \eta_*$ *for all* $t \ge 1$*). Then* $v_t \neq 0$ *for every* t*, and:*

(a) $v_t = (I - \gamma H)^t v_0$*, i.e.* $q_k^\top v_t = (1 - \gamma\lambda_k)^t\, q_k^\top v_0$ *for all* k*, and* $\mu_{\mathrm{eff}}(\theta_t) = \sum_k \lambda_k \pi_k(t)$ *with* $\pi_k(t) \propto (1 - \gamma\lambda_k)^{2t}(q_k^\top v_0)^2$*.*

(b) (Monotonicity) $t \mapsto \mu_{\mathrm{eff}}(\theta_t)$ *is non-increasing.*

(c) (Alignment) If $Pv_0 \neq 0$ *(true for Lebesgue-almost every* v_0*), then with* $\rho_{\mathrm{gap}} = (1 - \gamma\lambda')/(1 - \gamma\lambda_{\min}) \in [0, 1)$*,*

$$\frac{\|v_t - Pv_t\|}{\|Pv_t\|} \;\le\; \rho_{\mathrm{gap}}^{t}\, \frac{\|v_0 - Pv_0\|}{\|Pv_0\|},$$

so that $v_t / \|v_t\|$ *converges to* $E_{\min}$ *at the geometric rate* ρ_{gap}*, and*

$$0 \;\le\; \mu_{\mathrm{eff}}(\theta_t) - \lambda_{\min} \;\le\; (\lambda_1 - \lambda_{\min})\, \rho_{\mathrm{gap}}^{2t}\, \frac{\|v_0 - Pv_0\|^2}{\|Pv_0\|^2} \;\xrightarrow[t\to\infty]{}\; 0. \tag{B2.29}$$

(d) (Invariant subspaces) If S *is a linear subspace with* $HS \subseteq S$ *and* $v_0 \in S$*, then* $v_t \in S$ *for all* t*, and (a)–(c) hold with* H *replaced by its restriction* $H|_S$*: the sums run over the eigenvectors* $q_k \in S$ *only,* $\lambda_{\min}$ *is replaced by* $\lambda_{\min}(H|_S) = \min\{\lambda_k : q_k \in S\}$*,* $E_{\min}$ *and* P *by the corresponding eigenspace within* S *and its orthogonal projector,* λ' *by the second smallest distinct eigenvalue of* $H|_S$*, and the hypothesis* $Pv_0 \neq 0$ *by its analogue for the restricted projector.*

[proof: Appendix B2 of the full report, p. 106]

The alignment time is of order $1/(\gamma(\lambda' - \lambda_{\min}))$ iterations, which can exceed the horizon of an experiment when the spectrum is dense near $\lambda_{\min}$; monotonicity (b) is then the robust prediction, as Remark B2.18 (scope of Proposition B2.17) explains. For SAGA, the mean error follows gradient descent exactly, while the second moment adds the estimator noise accumulated along the trajectory.

Lemma B2.21 (SAGA on a quadratic: exact moment recursions)**.** *Let* $f_i(\theta) = \frac{1}{2}\theta^\top H_i \theta - b_i^\top \theta + c_i$ *with* H_i *symmetric,* $0 \preccurlyeq H_i \preccurlyeq R^2 I$*, and* $H = \frac{1}{n}\sum_i H_i \succcurlyeq \mu I$*. Consider SAGA with step* γ *and memory* $z_i^{(t)}$ *as in the book, write* $v_t = \theta_t - \eta_*$*, and define the memory errors* $e_i^{(t)} = z_i^{(t)} - f_i'(\eta_*)$ *and* $\bar e^{(t)} = \frac{1}{n}\sum_i e_i^{(t)}$*. Let* $i(t)$ *be uniform on* $\{1, \dots, n\}$ *and independent of the past* $\mathcal{F}_{t-1} = \sigma(i(1), \dots, i(t-1))$*.*

(a) (State recursion) $v_t = (I - \gamma H)v_{t-1} + \gamma\xi_t$ *with*

$$\xi_t = \big(e_{i(t)}^{(t-1)} - \bar e^{(t-1)}\big) - \big(H_{i(t)} - H\big)v_{t-1}, \qquad \mathbb{E}_{i(t)}[\xi_t] = 0,$$

and $e_{i(t)}^{(t)} = H_{i(t)} v_{t-1}$*,* $e_j^{(t)} = e_j^{(t-1)}$ *for* $j \neq i(t)$*. Moreover* $\xi_t = F'(\theta_{t-1}) - \omega_t$ *is minus the error of the SAGA gradient estimator* ω_t*.*

(b) (Mean) $\mathbb{E}[v_t] = (I - \gamma H)^t v_0$*: the mean error follows exactly the gradient descent recursion.*

(c) (Second moment) With $C_t = \mathbb{E}_{i(t)}[\xi_t \xi_t^\top] = \frac{1}{n}\sum_{i=1}^n g_i^{(t)} g_i^{(t)\top}$*, where* $g_i^{(t)} = (e_i^{(t-1)} - \bar e^{(t-1)}) - (H_i - H)v_{t-1}$ *(so that* C_t *is the conditional covariance of* ω_t *given* $\mathcal{F}_{t-1}$*),*

$$\mathbb{E}[v_t v_t^\top] = (I - \gamma H)\,\mathbb{E}[v_{t-1} v_{t-1}^\top]\,(I - \gamma H) + \gamma^2 \mathbb{E}[C_t],$$

and therefore, for every eigenpair (λ_k, q_k) *of* H,

$$\mathbb{E}\big[(q_k^\top v_t)^2\big] = (1-\gamma\lambda_k)^{2t}(q_k^\top v_0)^2 + \gamma^2 \sum_{s=1}^{t} (1-\gamma\lambda_k)^{2(t-s)}\, q_k^\top \mathbb{E}[C_s]\, q_k. \tag{B2.30}$$

[proof: Appendix B2 of the full report, p. 108]

Assuming that the noise covariance decays at the certified rate of Proposition 5.9, which is slower than every mode of the mean dynamics, a quasi-stationary heuristic predicted that the error of SAGA stays spread over the directions fed by the noise, with $\mu_{\text{eff}}(\theta_t)$ fluctuating above $\lambda_{\min}(H_*)$ (prediction (P2) of Section B2.4; (P1) predicts that along gradient descent $\mu_{\text{eff}}(\theta_t)$ decreases to $\lambda_{\min}(H_*)$ when the features span $\mathbb{R}^d$). In nine runs on logistic regression ($n = 1000$, $d = 50$, one seed; SAGA with $\gamma = \frac{1}{4R^2}$, gradient descent with step $1/L$), (P1) is confirmed up to the speed of convergence, whereas (P2) is confirmed only in the four runs with $9n\lambda_{\min}(H_*)/(16R^2) = \frac{9}{4}n\gamma\lambda_{\min}(H_*) > 1$ ($\kappa = 10^2$, all α; $\kappa = 10^3$, $\alpha = 0$). It is refuted for $\kappa = 10^4$, $\alpha \in \{0, 1\}$, where SAGA, started from an error that already carries 89–98% of its squared norm on the flattest direction of H_*, keeps and sharpens this alignment, like gradient descent; the three other runs are intermediate. Thus (P2) holds only for κ at most of the order of n, and not in every such run. Remark B2.22 (outcome of prediction (P2)) reports the nine runs and explains the alignment heuristically; Section 5.5 discusses them.

How the effective constants enter the bound. Keeping I4 and I2 as identities in the proof of Proposition 5.9 gives the a-posteriori one-step factors of Proposition B3.19 (a-posteriori rates with effective constants), rederived as Proposition B2.23 (trajectory-dependent one-step contraction). As noted after Proposition B3.19, $\mu_{\text{eff}}(\theta_{t-1})$ enters them linearly, whereas, at the fixed step $\gamma = \frac{1}{4R^2}$, $L_{\text{eff}} < R^2$ improves the parameter rate by a factor at most $\frac{8}{3}$; moreover, once μ_{eff} exceeds $16R^2/(9n)$, the memory rate $\frac{1}{3n}$ binds and no further gain in curvature shows in the bound, as Remark B2.24 (what each constant can explain) details. In order B of Proposition B3.11 (attribution of the gap), undoing I4 and I2 replaces μ and R^2 by μ_{eff} and L_{eff}; for $\kappa \geq 10^4$ ($n = 1000$, $d = 50$, $\alpha \in \{0, 1, 2\}$, 5 seeds) these two steps carry 77–96% of the logarithmic gap (Section 5.3). In order A, which keeps the minimum until the end and whose first two steps give the two one-step factors of Proposition B3.19, I4 and I2 carry 59–96% of the gap in the same runs (Section 5.5).

Known and new. That algorithms meet the local curvature is classical: the local strong-convexity constant tends to $\lambda_{\min}(H_*)$ [9, Section 5.2.3], averaged SGD adapts to it on logistic regression [8] through self-concordance [6], $\mu_{\text{eff}}(\theta)$ is the constant of the restricted secant inequality at θ [30], and L_{eff} is a trajectory-wise version of the expected-smoothness constant [15, 16]. To our knowledge, the precise form of Proposition B2.12 (the global strong convexity constant is λ) and Corollary B2.15 (effective curvature versus curvature at the optimum), and the comparison of the directions selected by gradient descent and SAGA, whose heuristic prediction the experiments partly refute, are not in the sources we reviewed in this form; the integral form, the ranges and the directional limits (Lemma B2.5, Lemma B2.6 and Proposition B2.8) are elementary consequences of standard facts, collected here for the trajectory constants.

4.3 Preconditioned SAGA

Hypothesis B1 runs SAGA in a fixed quadratic metric to see which steps of the proof of Proposition 5.9 are too loose, and asks whether the metric improves the algorithm or only its analysis. Let P be a fixed symmetric positive definite matrix, $\text{cond}(P) = \lambda_{\max}(P)/\lambda_{\min}(P)$ its condition number,

$\|v\|_P = (v^\top P v)^{1/2}$ the norm of parameters and $\|g\|_{P^{-1}} = (g^\top P^{-1} g)^{1/2}$ the dual norm of gradients and memories. The algorithm of Definition B1.3 (P-SAGA: SAGA in the metric of P) keeps the direction ω_t and the memory of SAGA (3) and, by Lemma B1.4 (closed form, unbiasedness, Bregman form), steps $\theta_t = \theta_{t-1} - \gamma P^{-1}\omega_t$; it is Bregman-SAGA [11] with the reference function $h(\theta) = \frac{1}{2}\theta^\top P \theta$. A linear change of variables reduces it to SAGA:

Proposition B1.5 (P-SAGA as SAGA on a transformed problem). *Let $u := P^{1/2}\theta$, $\tilde{f}_i(u) := f_i(P^{-1/2}u)$ and $\tilde{F} := \frac{1}{n}\sum_i \tilde{f}_i = F \circ P^{-1/2}$. Then:*

(i) $\nabla \tilde{f}_i(u) = P^{-1/2} f_i'(P^{-1/2}u)$ *and, when f_i is C^2,* $\nabla^2 \tilde{f}_i(u) = P^{-1/2} f_i''(P^{-1/2}u) P^{-1/2}$*; the same formulas hold for $(\tilde{F}, F)$.*

(ii) $\tilde{F}$ is strongly convex and $\tilde{u}_* := P^{1/2}\eta_*$ is its unique minimizer.

(iii) Let $(\theta_t, z^{(t)})_{t\geq 0}$ be generated by P-SAGA (Definition B1.3) with step γ and index sequence $(i(t))_{t\geq 1}$, and let $u_t := P^{1/2}\theta_t$, $\tilde{z}_i^{(t)} := P^{-1/2} z_i^{(t)}$. Then $(u_t, \tilde{z}^{(t)})_{t\geq 0}$ is generated by Euclidean SAGA ($P = I$) on $(\tilde{f}_i)_{i\leq n}$ with the same *step γ and the* same *index sequence, started from $(u_0, \tilde{z}^{(0)})$.*

(iv) For all t and i, $\|u_t - \tilde{u}_*\|^2 = \|\theta_t - \eta_*\|_P^2$ and $\left\|\tilde{z}_i^{(t)} - \nabla \tilde{f}_i(\tilde{u}_*)\right\|^2 = \left\|z_i^{(t)} - f_i'(\eta_*)\right\|_{P^{-1}}^2$; in particular $z_i^{(0)} = f_i'(\theta_0)$ for all i iff $\tilde{z}_i^{(0)} = \nabla \tilde{f}_i(u_0)$ for all i.

[proof: Appendix B1 of the full report, p. 119]

Every step of the proof of Proposition 5.9 thus holds for P-SAGA, in the norms $\|\cdot\|_P$ and $\|\cdot\|_{P^{-1}}$ and with the constants of the transformed functions, written here for C^2 functions (for C^1 functions, see Remark B1.10):

Definition B1.7 (relative constants and relative condition number). *We set*

$$R_P^2 := \max_{1\leq i\leq n} \sup_{\theta\in\mathbb{R}^d} \lambda_{\max}\big(P^{-1/2} f_i''(\theta) P^{-1/2}\big),$$
$$\mu_P := \inf_{\theta\in\mathbb{R}^d} \lambda_{\min}\big(P^{-1/2} F''(\theta) P^{-1/2}\big), \qquad \kappa_P := \frac{R_P^2}{\mu_P}. \tag{B1.6}$$

By Lemma B1.6, every $\tilde{f}_i$ of Proposition B1.5 is R_P^2-smooth and $\tilde{F}$ is μ_P-strongly convex; in the terminology of [19, Def. 1] and [26], f_i is R_P^2-smooth and F is μ_P-strongly convex relative to *the reference function $h = \frac{1}{2}\|\cdot\|_P^2$ (whose Hessian is P). For $P = I$ these are the usual constants, and R^2, μ, κ always denote the Euclidean ($P = I$) constants.* [Appendix B1 of the full report, p. 121]

Proposition 5.9 then transfers verbatim, with these constants in place of (R^2, μ):

Corollary B1.12 (Proposition 5.9 transferred to P-SAGA). *Let P be SPD and let R_P^2, μ_P be the constants of (B1.6) (or any constants such that each $\tilde{f}_i$ of Proposition B1.5 is R_P^2-smooth and $\tilde{F}$ is μ_P-strongly convex). Run P-SAGA (Definition B1.3) with $\gamma = 1/(4R_P^2)$ and define*

$$V_t^P := \|\theta_t - \eta_*\|_P^2 + 3\gamma^2 \sum_{i=1}^n \left\|z_i^{(t)} - f_i'(\eta_*)\right\|_{P^{-1}}^2,$$
$$\rho_P := 1 - \min\left\{\frac{1}{3n}, \frac{3\mu_P}{16 R_P^2}\right\} = 1 - \min\left\{\frac{1}{3n}, \frac{3}{16\kappa_P}\right\}. \tag{B1.11}$$

Then for every $t \geq 1$,

$$\mathbb{E}_{i(t)}\left[V_t^P\right] \leq \rho_P\, V_{t-1}^P, \tag{B1.12}$$

and if $z_i^{(0)} = f_i'(\theta_0)$ for all i,

$$\mathbb{E}\,\|\theta_t - \eta_*\|_P^2 \leq \rho_P^t\Big(1 + \frac{n}{4}\Big)\,\|\theta_0 - \eta_*\|_P^2\,. \tag{B1.13}$$

Consequently

$$\mathbb{E}\,\|\theta_t - \eta_*\|^2 \leq \mathrm{cond}(P)\,\rho_P^t\Big(1 + \frac{n}{4}\Big)\,\|\theta_0 - \eta_*\|^2$$

(the condition number of P enters the constant, not the rate), and $t \geq \max\{3n, \frac{16}{3}\kappa_P\}\log\frac{2n}{\varepsilon}$ iterations guarantee $\mathbb{E}\,\|\theta_t - \eta_*\|_P^2 \leq \varepsilon\,\|\theta_0 - \eta_*\|_P^2$, i.e. a running time of order $c_P\,(3n + \frac{16}{3}\kappa_P)\log\frac{2n}{\varepsilon}$, where c_P is the cost of one iteration (Section B1.5). [proof: Appendix B1 of the full report, p. 122]

The metric acts on the rate only through κ_P; $\mathrm{cond}(P)$ enters only the constant of the Euclidean bound. By Lemma B1.8 (comparison with the Euclidean constants), $\kappa/\,\mathrm{cond}(P) \leq \kappa_P \leq \kappa\,\mathrm{cond}(P)$, so that a fixed metric changes the certified rate by at most the factor $\mathrm{cond}(P)$, in either direction. In the proof written in the P-metric (Section B1.3.1), I1 and I3 change only their norm, I2 and I4 their norm and their constant (R_P^2, μ_P), I5 depends on P only through $3\mu_P/(16R_P^2)$, and I6 through both norms and R_P^4. The tightness ratios of I2 and I4 in the P-geometry (Definition B1.14) are $r_{\mathrm{coco}}^P(t) = L_{\mathrm{eff}}^P(\theta_{t-1})/R_P^2$ and $r_{\mathrm{sc}}^P(t) = \mu_P/\mu_{\mathrm{eff}}^P(\theta_{t-1})$, with the effective constants $\mu_{\mathrm{eff}}^P(\theta) = \langle F'(\theta), \theta - \eta_*\rangle\,/\,\|\theta - \eta_*\|_P^2$ and $L_{\mathrm{eff}}^P(\theta) = \frac{1}{n}\sum_i \|f_i'(\theta) - f_i'(\eta_*)\|_{P^{-1}}^2\,/\,\langle F'(\theta), \theta - \eta_*\rangle$ $(\theta \neq \eta_*)$. By Remark B1.16 (effective one-step contraction), if I2 and I4 are replaced by the identities that define these constants, the parameter part of the one-step bound (B1.28) contracts by a factor between $1 - \mu_{\mathrm{eff}}^P/(2R_P^2)$ and $1 - 3\mu_{\mathrm{eff}}^P/(16R_P^2)$, with μ_{eff}^P evaluated at θ_{t-1}: the rate seen by the proof is governed by $R_P^2/\mu_{\mathrm{eff}}^P(\theta_{t-1})$.

Global constants of logistic regression. For ℓ_2-regularized logistic regression, μ_P is explicit and R_P^2 is bounded on both sides ($\sigma_1 \geq \cdots \geq \sigma_d$ denote the eigenvalues of $\hat{\Sigma}$, not the per-sample curvatures $\sigma_i'(\theta)$ of the lemma):

Lemma B1.11 (relative constants of logistic regression)**.** *Let $f_i(\theta) = \ell(y_i x_i^\top\theta) + \frac{\lambda}{2}\,\|\theta\|^2$ with $\ell(u) = \log(1 + e^{-u})$, $y_i \in \{-1, 1\}$, $x_i \in \mathbb{R}^d$, $\lambda > 0$, and let $\sigma(u) = (1 + e^{-u})^{-1}$. Then $\ell'(u) = -\sigma(-u)$ and $\ell''(u) = \sigma'(u) = \sigma(u)\sigma(-u) = \big(2 + e^u + e^{-u}\big)^{-1} \in (0, \frac{1}{4}]$, which equals $\frac{1}{4}$ at $u = 0$ and is strictly decreasing in $|u|$. Writing $\sigma_i'(\theta) := \sigma'(y_i x_i^\top\theta)$,*

$$f_i'(\theta) = \ell'(y_i x_i^\top\theta)\,y_i x_i + \lambda\theta, \qquad f_i''(\theta) = \sigma_i'(\theta)\,x_i x_i^\top + \lambda I, \qquad F''(\theta) = \frac{1}{n}\sum_{i=1}^n \sigma_i'(\theta)\,x_i x_i^\top + \lambda I. \tag{B1.8}$$

For every SPD P,

$$\max\left\{\frac{1}{4}\max_i \|x_i\|_{P^{-1}}^2\,,\ \frac{\max_i \|x_i\|^2/4 + \lambda}{\lambda_{\max}(P)}\right\} \ \leq\ R_P^2\ \leq\ \frac{1}{4}\max_i \|x_i\|_{P^{-1}}^2 + \frac{\lambda}{\lambda_{\min}(P)}, \tag{B1.9}$$

and more precisely $R_P^2 \geq \frac{1}{4}\,\|x_i\|_{P^{-1}}^2 + \lambda\,\frac{x_i^\top P^{-2} x_i}{x_i^\top P^{-1} x_i}$ for every i with $x_i \neq 0$;

$$\mu_P = \lambda\,\lambda_{\min}(P^{-1}) = \frac{\lambda}{\lambda_{\max}(P)}. \tag{B1.10}$$

In particular, for $P = I$: $R^2 := \frac{1}{4}\max_i \|x_i\|^2 + \lambda$ is a valid smoothness constant of every f_i, $\mu = \lambda$, and $\kappa = R^2/\mu = \max_i \|x_i\|^2/(4\lambda) + 1$; these are the Euclidean constants used in the whole project. [proof: Appendix B1 of the full report, p. 122]

Comparing the lower bound on R_P^2 with μ_P gives the central negative result:

Proposition B1.21 (no free lunch for global constants). *For ℓ_2-regularized logistic regression and* every *SPD matrix P,*

$$\kappa_P \;\geq\; \frac{\max_i \|x_i\|^2}{4\lambda} + 1 = \kappa, \tag{B1.33}$$

with κ the Euclidean condition number (B1.30); consequently $\rho_P \geq \rho_I$ for every P. In particular, for $P = \hat{\Sigma} + \lambda I$, $\kappa \leq \kappa_P \leq \kappa \, \frac{\sigma_1+\lambda}{\sigma_d+\lambda}$. [proof: Appendix B1 of the full report, p. 130]

The supremum in R_P^2 is attained on the decision boundary of a sample, where its data curvature is $\frac{1}{4}x_i x_i^\top$, whereas the infimum in μ_P is approached far from all of them, where $F'' \to \lambda I$ is isotropic; no single metric fits both, and with global constants the transferred Proposition 5.9 never certifies a faster rate than for $P = I$ on this problem. For the square loss, by contrast, $P = \hat{\Sigma} + \lambda I$ brings κ_P into $[d_\infty(\lambda), d_\infty(\lambda) + 1]$ (the ridge setting of [14]; Section B1.4). Here, with X the matrix of rows $x_i^\top$, $\mathrm{lev}_i(\lambda) = x_i^\top (X^\top X + n\lambda I)^{-1} x_i$ are the ridge leverage scores [13, 7], $d_\infty(\lambda) = n \max_i \mathrm{lev}_i(\lambda)$ and $d_{\mathrm{eff}}(\lambda) = \sum_i \mathrm{lev}_i(\lambda) = \mathrm{tr}(\hat{\Sigma}(\hat{\Sigma}+\lambda I)^{-1})$; by Lemma B1.17 (leverage scores), $d_{\mathrm{eff}}(\lambda) \leq \min\{d_\infty(\lambda), d\}$ and $d_\infty(\lambda) < n$.

Trajectory-dependent constants. For logistic regression, any benefit of a metric must therefore come from the trajectory-dependent constants, which are explicit in terms of the per-sample curvatures averaged on the segment $[\eta_*, \theta]$:

Proposition B1.22 (effective strong convexity from segment curvatures). *For the logistic regression of Lemma B1.11, every $\theta \neq \eta_*$, $\delta := \theta - \eta_*$, and $\bar{\sigma}_i'(\theta) := \int_0^1 \sigma'\big(y_i x_i^\top (\eta_* + s\delta)\big)\, ds \in (0, \frac{1}{4}]$,*

$$f_i'(\theta) - f_i'(\eta_*) = \bar{\sigma}_i'(\theta)\,(x_i^\top \delta)\, x_i + \lambda\delta, \qquad \big\langle F'(\theta), \theta - \eta_* \big\rangle = \frac{1}{n}\sum_{i=1}^n \bar{\sigma}_i'(\theta)\,(x_i^\top \delta)^2 + \lambda\,\|\delta\|^2 . \tag{B1.34}$$

Consequently, with $m(\theta) := \min_i \bar{\sigma}_i'(\theta) \;\geq\; \min_i \sigma'\big(\max\{|x_i^\top \eta_*|, |x_i^\top \theta|\}\big) > 0$:

(i) for $P = \hat{\Sigma} + \lambda I$, the effective strong convexity $\mu_{\mathrm{eff}}^P(\theta) := \langle F'(\theta), \delta\rangle \,/\, \|\delta\|_P^2$ is the convex combination

$$\mu_{\mathrm{eff}}^P(\theta) = \frac{\frac{1}{n}\sum_i \bar{\sigma}_i'(\theta)(x_i^\top \delta)^2 + \lambda\,\|\delta\|^2}{\frac{1}{n}\sum_i (x_i^\top \delta)^2 + \lambda\,\|\delta\|^2} \;\in\; [\,m(\theta),\, 1\,] \tag{B1.35}$$

of the numbers $\bar{\sigma}_1'(\theta), \dots, \bar{\sigma}_n'(\theta), 1$, and more precisely

$$\mu_{\mathrm{eff}}^P(\theta) \;\geq\; \frac{m(\theta)\,\sigma_1 + \lambda}{\sigma_1 + \lambda} \;\geq\; \max\Big\{m(\theta),\; \frac{\lambda}{\sigma_1+\lambda}\Big\} = \max\{m(\theta), \mu_P\}; \tag{B1.36}$$

(ii) for $P = I$, $\mu_{\mathrm{eff}}(\theta) = \frac{1}{n}\sum_i \bar{\sigma}_i'(\theta)\,(x_i^\top \delta/\,\|\delta\|)^2 + \lambda \in [\,m(\theta)\sigma_d + \lambda,\; \sigma_1/4 + \lambda\,]$.

[proof: Appendix B1 of the full report, p. 130]

A sample with a large margin makes $m(\theta)$ small ($\sigma'(u) \approx e^{-|u|}$) but has a small weight in (B1.35) unless $\theta - \eta_*$ is aligned with it, so that $\mu_{\mathrm{eff}}^P(\theta)$ is typically much larger than $m(\theta)$; the global version of (B1.36) is only $\mu_P = \lambda/(\sigma_1 + \lambda)$, because $\inf_\theta \min_i \sigma'(y_i x_i^\top \theta) = 0$. The smoothness side is controlled by the leverage scores, by a multiple of the effective dimension when they are incoherent:

Assumption B1.18 (coherence of the leverage scores). *There is $C \geq 1$ such that the ridge leverage scores (B1.29) satisfy $\max_i \mathrm{lev}_i(\lambda) \leq C\, d_{\mathrm{eff}}(\lambda)/n$, i.e. $d_\infty(\lambda) \leq C\, d_{\mathrm{eff}}(\lambda)$: no data point has a leverage score larger than C times the average leverage score.* [Appendix B1 of the full report, p. 128]

Since $\|x_i\|^2_{(\hat\Sigma+\lambda I)^{-1}} = n\,\mathrm{lev}_i(\lambda)$ (Lemma B1.17(i)), Lemma B1.11 gives, for the statistical metric:

Proposition B1.20 (global constants of the statistical preconditioner). *For $P = \hat\Sigma + \lambda I$,*

$$\frac{d_\infty(\lambda)}{4} \;\le\; R_P^2 \;\le\; \frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d+\lambda} \;\le\; \frac{d_\infty(\lambda)}{4} + 1 \;\le\; \frac{C\,d_{\mathrm{eff}}(\lambda)}{4} + 1, \tag{B1.31}$$

the last inequality under Assumption B1.18, and

$$\mu_P = \frac{\lambda}{\sigma_1+\lambda}, \qquad \kappa_P = \frac{R_P^2}{\mu_P} \in \Big[\frac{d_\infty(\lambda)}{4}\cdot\frac{\sigma_1+\lambda}{\lambda},\; \Big(\frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d+\lambda}\Big)\frac{\sigma_1+\lambda}{\lambda}\Big]. \tag{B1.32}$$

[proof: Appendix B1 of the full report, p. 129]

Dividing (B1.31) by (B1.36) bounds the effective relative condition number:

Corollary B1.23 (leverage-score bound along trajectories). *Let $\kappa^P_{\mathrm{eff}}(\theta) := R_P^2/\mu^P_{\mathrm{eff}}(\theta)$ be the effective relative condition number at θ (the quantity governing the effective contraction (B1.28)). Then, for $P = \hat\Sigma + \lambda I$ and $m(\theta)$ as in Proposition B1.22,*

$$\kappa^P_{\mathrm{eff}}(\theta) \;\le\; \Big(\frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d+\lambda}\Big)\frac{\sigma_1+\lambda}{m(\theta)\sigma_1+\lambda} \;\le\; \frac{d_\infty(\lambda)/4+1}{m(\theta)} \;\le\; \frac{C\,d_{\mathrm{eff}}(\lambda)/4+1}{m(\theta)}, \tag{B1.37}$$

the last inequality under Assumption B1.18, whereas for $P = I$ the same argument only gives

$$\frac{\max_i \|x_i\|^2/4+\lambda}{\sigma_1/4+\lambda} \;\le\; \kappa_{\mathrm{eff}}(\theta) \;\le\; \frac{\max_i \|x_i\|^2/4+\lambda}{m(\theta)\sigma_d+\lambda}, \qquad \textit{with} \qquad \frac{\max_i \|x_i\|^2/4+\lambda}{\sigma_1/4+\lambda} \;\ge\; \frac{\mathrm{tr}\,\hat\Sigma + 4\lambda}{\sigma_1+4\lambda}. \tag{B1.38}$$

[proof: Appendix B1 of the full report, p. 131]

In the last two bounds of (B1.37), the eigenvalues of $\hat\Sigma$ appear only through $d_\infty(\lambda) \le C\,d_{\mathrm{eff}}(\lambda)$, and $d_{\mathrm{eff}}(\lambda)$ counts the directions not killed by the regularization, whatever the dynamic range of the spectrum. The price is the smallest segment curvature $m(\theta)$, and on the instances of Section 5.6 it dominates. There, as $\theta \to \eta_*$, $m(\theta) \to \min_i w_i$ with $w_i = \sigma'(y_i x_i^\top \eta_*)$, and $\min_i w_i = 7.4\cdot 10^{-3}$ ($\alpha = 1$) and $6.1\cdot 10^{-3}$ ($\alpha = 2$), so that the first bound in (B1.37) tends to $1.7\cdot 10^3$ and $1.2\cdot 10^3$: above $\kappa = 10^3$, and above the upper bound in (B1.38) for $P = I$ with the same $m(\theta)$ (968 and 999). On these instances this rigorous bound therefore predicts no gain of $\hat\Sigma + \lambda I$.

Jacobi and Newton metrics; local constants. The Jacobi metric $P_{\mathrm{J}} = \mathrm{diag}(\hat\Sigma) + \lambda I$ costs $O(d)$ per iteration, like $P = I$, whereas a dense P costs $O(d^2)$ per iteration and a precomputation of order $nd^2 + d^3$ (Section B1.5). With $\hat\Sigma_{jj} > 0$ the diagonal entries of $\hat\Sigma$, x_{ij} the entries of x_i and $C_{\mathrm{corr}} = \mathrm{diag}(\hat\Sigma)^{-1/2}\hat\Sigma\,\mathrm{diag}(\hat\Sigma)^{-1/2}$ the correlation matrix of the features, the constants of P_{J} compare with those of $\hat\Sigma + \lambda I$ as follows:

Proposition B1.19 (constants of the Jacobi preconditioner). *(i)* $\|x_i\|^2_{P_{\mathrm{J}}^{-1}} = \sum_{j=1}^d \frac{x_{ij}^2}{\hat\Sigma_{jj}+\lambda}$ *and*

$$\frac{1}{n}\sum_i \|x_i\|^2_{P_{\mathrm{J}}^{-1}} = \sum_{j=1}^d \frac{\hat\Sigma_{jj}}{\hat\Sigma_{jj}+\lambda} =: d_{\mathrm{eff}}^{\mathrm{diag}}(\lambda) \;\ge\; d_{\mathrm{eff}}(\lambda).$$

(ii) $R^2_{P_{\mathrm{J}}} \le \frac14 \max_i \|x_i\|^2_{P_{\mathrm{J}}^{-1}} + \frac{\lambda}{\min_j \hat\Sigma_{jj}+\lambda}$ *and* $\mu_{P_{\mathrm{J}}} = \frac{\lambda}{\max_j \hat\Sigma_{jj}+\lambda}$.

(iii) $\lambda_{\min}(C_{\mathrm{corr}})\,P_{\mathrm{J}} \preceq \hat{\Sigma} + \lambda I \preceq \lambda_{\max}(C_{\mathrm{corr}})\,P_{\mathrm{J}}$; *consequently* $\lambda_{\min}(C_{\mathrm{corr}})\,n\,\mathrm{lev}_i(\lambda) \le \|x_i\|^2_{P_{\mathrm{J}}^{-1}} \le \lambda_{\max}(C_{\mathrm{corr}})\,n\,\mathrm{lev}_i(\lambda)$ *for all* i.

[proof: Appendix B1 of the full report, p. 129]

By (iii), the constants of P_{J} are those of $\hat{\Sigma}+\lambda I$ up to the factor $\mathrm{cond}(C_{\mathrm{corr}})$, in either direction: P_{J} removes the scale anisotropy of the features but not their correlations, and nothing orders it with respect to $P = I$. The oracle Newton metric $H = F''(\eta_*)$, which requires η_*, uses the curvatures $w_i \in (0, \frac{1}{4}]$ at the optimum: with $W = \mathrm{diag}(w_1, \dots, w_n)$ and $\hat{\Sigma}_w = \frac{1}{n}X^\top W X$ (eigenvalues $\sigma_1^w \ge \cdots \ge \sigma_d^w$), $H = \hat{\Sigma}_w + \lambda I$, and its constants involve the leverage scores $\mathrm{lev}_i^w(\lambda) = w_i x_i^\top (X^\top W X + n\lambda I)^{-1} x_i$ of the weighted design and $d_\infty^w(\lambda) = n \max_i \mathrm{lev}_i^w(\lambda)$:

Proposition B1.26 (constants in the Newton metric). *For* $P = H = F''(\eta_*)$:

(i) $\|x_i\|^2_{H^{-1}} = n\,\mathrm{lev}_i^w(\lambda)/w_i$, *hence*

$$\max_i \frac{n\,\mathrm{lev}_i^w(\lambda)}{4w_i} \;\le\; R_H^2 \;\le\; \max_i \frac{n\,\mathrm{lev}_i^w(\lambda)}{4w_i} + \frac{\lambda}{\sigma_d^w + \lambda} \;\le\; \frac{d_\infty^w(\lambda)}{4\min_i w_i} + 1, \qquad \mu_H = \frac{\lambda}{\sigma_1^w + \lambda}. \tag{B1.42}$$

(ii) *For* $\theta \ne \eta_*$, $\delta := \theta - \eta_*$ *and* $\bar{\sigma}_i'(\theta)$ *as in Proposition B1.22,*

$$\mu_{\mathrm{eff}}^H(\theta) = \frac{\frac{1}{n}\sum_i \bar{\sigma}_i'(\theta)(x_i^\top\delta)^2 + \lambda\|\delta\|^2}{\frac{1}{n}\sum_i w_i(x_i^\top\delta)^2 + \lambda\|\delta\|^2} \;\in\; \Big[\min\Big\{1, \min_i \frac{\bar{\sigma}_i'(\theta)}{w_i}\Big\},\; \max\Big\{1, \max_i \frac{\bar{\sigma}_i'(\theta)}{w_i}\Big\}\Big], \tag{B1.43}$$

a convex combination of $\bar{\sigma}_1'/w_1, \dots, \bar{\sigma}_n'/w_n, 1$; *in particular* $\mu_{\mathrm{eff}}^H(\theta) \to 1$ *as* $\theta \to \eta_*$, *and* $\mu_{\mathrm{eff}}^H(\theta) \ge \mu_H$ *always.*

[proof: Appendix B1 of the full report, p. 133]

The Newton metric makes strong convexity perfect near the optimum, but R_H^2 still pays the supremum over θ: sample i enters it with its largest curvature $\frac{1}{4}$ in the numerator and its curvature w_i at the optimum in the metric, hence the factor $1/w_i$ in (B1.42). Near the optimum, the realized smoothness constant does not pay this factor, and the realized strong convexity is a local curvature:

Lemma B1.27 (limits of the effective constants at η_*). *Let the* f_i *be* C^2, $H_i := f_i''(\eta_*)$, $H := F''(\eta_*) = \frac{1}{n}\sum_i H_i \succ 0$, *and let* $\theta \to \eta_*$ *with* $\delta := \theta - \eta_*$ *satisfying* $\delta/\|\delta\| \to v$, $\|v\| = 1$. *Then, for every SPD* P,

$$\begin{aligned} L_{\mathrm{eff}}^P(\theta) &\longrightarrow \frac{\frac{1}{n}\sum_i v^\top H_i P^{-1} H_i v}{v^\top H v}, \\ \mu_{\mathrm{eff}}^P(\theta) &\longrightarrow \frac{v^\top H v}{v^\top P v} \in \big[\lambda_{\min}(P^{-1/2}HP^{-1/2}), \lambda_{\max}(P^{-1/2}HP^{-1/2})\big]. \end{aligned} \tag{B1.44}$$

Moreover $\frac{1}{n}\sum_i H_i P^{-1} H_i \succeq HP^{-1}H$, *so the limit of* L_{eff}^P *is at least* $\lambda_{\min}(P^{-1/2}HP^{-1/2})$, *and for* $P = H$

$$\lim L_{\mathrm{eff}}^H(\theta) = 1 + \frac{\frac{1}{n}\sum_i \big\|H^{-1/2}(H_i - H)v\big\|^2}{\big\|H^{1/2}v\big\|^2} \;\ge\; 1, \tag{B1.45}$$

with equality iff $H_i v = Hv$ *for all* i *(homogeneous per-sample curvature in the direction* v*). For the logistic regression of Lemma B1.11, in addition,* $\lim L_{\mathrm{eff}}^H(\theta) \le \max\{d_\infty^w(\lambda), 2\}$ *for*

$P = H$, $\lim L_{\text{eff}}^P(\theta) \le d_\infty(\lambda)/4 + \frac{3}{2}$ *for* $P = \hat\Sigma + \lambda I$, *and, for* $P = I$, *with* $c_i := x_i^\top v$ *and* $B_v := \frac{1}{n}\sum_i w_i^2 c_i^2 \|x_i\|^2$,

$$\lim L_{\text{eff}}(\theta) = \frac{\lambda^2 + 2\lambda\, v^\top \hat\Sigma_w v + B_v}{\lambda + v^\top \hat\Sigma_w v}, \qquad (v^\top \hat\Sigma_w v)^2 \le B_v \le \frac{\max_i \|x_i\|^2}{4}\, v^\top \hat\Sigma_w v = (\kappa - 1)\lambda\, v^\top \hat\Sigma_w v \tag{B1.46}$$

(this is the exact formula $L_{\text{eff}} = (\lambda^2 + 2\lambda A + B)/(\lambda + A)$ *of Lemma B2.10, in the limit, where its* A *and* B *become* $v^\top \hat\Sigma_w v$ *and* B_v*); in particular* $\lambda + v^\top \hat\Sigma_w v \le \lim L_{\text{eff}}(\theta) \le \lambda + (\kappa + 1)\, v^\top \hat\Sigma_w v$.

[proof: Appendix B1 of the full report, p. 134]

Only these local constants account for the gains observed in Section 5.6. Whatever the direction of the error, $\liminf_{\theta\to\eta_*} \mu_{\text{eff}}^P(\theta) \ge \lambda_{\min}(M_P)$ with $M_P = P^{-1/2} F''(\eta_*) P^{-1/2}$, so that $R_P^2/\lambda_{\min}(M_P)$ bounds $\limsup_{\theta\to\eta_*} \kappa_{\text{eff}}^P(\theta)$. For $P = \hat\Sigma + \lambda I$, $\lambda_{\min}(M_P)$ is the smallest over directions v of a weighted average of $w_1, \dots, w_n, 1$ (weights $(x_i^\top v)^2/n$ and $\lambda\|v\|^2$), not their minimum: on those instances it is 0.117 ($\alpha = 1$) and 0.109 ($\alpha = 2$), 16 and 18 times $\min_i w_i$, and $R_P^2/\lambda_{\min}(M_P) = 166$ and 103, against $R^2/\lambda_{\min}(F''(\eta_*)) = 576$ and 981 for $P = I$ (R_P^2 being the upper bound in (B1.9), as in the experiments); over passes 15–20 the measured $\kappa_{\text{eff}}^P(\theta_{t-1})$ are about 84 and 31, against 460 and 878 for $P = I$ (geometric means over states and seeds). These limits hold only as $\theta \to \eta_*$, and their use as rates along a whole run is heuristic; so used, the values 166 and 103 are below $\frac{9n}{16} = 562.5$ ($n = 1000$), and by Remark B1.24 (memory-bottleneck regime) the memory term $\frac{1}{3n}$ of the rate then binds. Through these limits, Section B1.6 also predicts which step is the loosest late in a run (prediction (P2), heuristic where it assumes the direction of the error): I2 for $P = I$, I4 with the global μ_P for the dense metrics (Table 4 in Appendix B1).

Known and new. Relative smoothness [26, 10], Bregman-SAGA (whose rate has the same structure, with constants $(\frac{1}{2}, \frac{1}{8})$ instead of $(\frac{1}{3}, \frac{3}{16})$ [11, Corollary 1]; Remark B1.13), the change of variables for a fixed metric [23, 14] and statistical preconditioning [19] are known. To our knowledge, Proposition B1.21 (no free lunch for global constants) and Corollary B1.23 (leverage-score bound along trajectories) are not in the sources we reviewed in this form; the first shows that, for logistic regression, the benefit of a fixed metric is entirely trajectory-dependent, and the second does not certify the gains observed in Section 5.6, which only the local constants of Lemma B1.27 (limits of the effective constants at η_*) account for, and only heuristically.

4.4 Lower bounds and the hard instances

Oracle classes, known bounds and hypothesis A1. Costs are counted in component gradients, a full gradient costing n; Table 1 collects the guarantees of the methods compared in this report and the main lower bounds. The lower bounds of the literature differ in how an algorithm may choose the indices of its queries and combine the answers; the following classes make these assumptions exact.

Definition A1.2 (linear-span protocol and classes (S1)–(S3))**.** *An algorithm follows the* linear-span protocol *(Nesterov's "black-box" assumption, [27] Section 2.1, relation (3.41) of [25], relation (1.2) of [18]) if, when it has queried the oracle at the pairs* (θ_{s-1}, i_s), $s = 1, \dots, t$, *its iterate satisfies*

$$\theta_t \in \theta_0 + \operatorname{span}\left\{ f'_{i_1}(\theta_0),\ f'_{i_2}(\theta_1),\ \dots,\ f'_{i_t}(\theta_{t-1}) \right\}. \tag{A1.1}$$

We distinguish three classes according to the index sequence (i_s)*:*

(S1) span-based with i.i.d. sampling*:* (i_s) *is i.i.d. from a fixed distribution* p *on* $\{1, \dots, n\}$ *(uniform in the project), independent of the oracle answers [25, 18]. This contains SGD, SAG,*

Finito/MISO, SDCA-without-duality, SAGA started from the zero memory $z^{(0)} = 0$, and, on block-separable problems, randomized block-coordinate methods. With the book's initialization $z_i^{(0)} = f_i'(\theta_0)$, SAGA is an (S1) method preceded by one full gradient at θ_0 *(n deterministic queries); Lemma A1.16(ii) and Corollary A1.21 treat this variant.*

(S2) span-based with an arbitrary index sequence*: (i_s) is any sequence, deterministic (cyclic, or n consecutive queries at the same point, which is a full gradient), random, or adaptive. This contains GD, Nesterov's accelerated gradient (AGD), cyclic incremental methods, and the hybrid methods SVRG, SARAH, L-SVRG, Katyusha and L-Katyusha, which mix i.i.d. single queries with full gradients.*

(S3) no span restriction*: arbitrary deterministic algorithms [1, 29] or arbitrary randomized algorithms [29], and the* oblivious canonical linear iterative *(CLI) class of [4], in which new iterates are linear combinations $\sum_j O_F(\theta^{(j)}; \alpha^{(j)})$ of oracle outputs (gradients possibly multiplied by fixed matrices, coordinate steps, full gradients) whose coefficients $\alpha^{(j)}$ are chosen* without looking at the oracle answers *(they may be random). Fixed step sizes, fixed momentum and fixed sampling schemes are allowed, line searches and quasi-Newton updates are not.*

[Appendix A1 of the full report, p. 143]

Lower bounds for (S1) therefore apply to SAGA, up to its initial sweep, but not to L-SVRG or L-Katyusha, whose full gradients are n queries at one point; those for (S2) apply to GD, AGD, SAGA, L-SVRG and L-Katyusha. On the chain constructions below, a span method reveals at most one new coordinate per query, so that span-based bounds require a dimension of the order of the number of queries they certify.

Hypothesis A1 (Section 1) asserts the bound $\Omega((n + \sqrt{n\kappa}) \log \frac{1}{\varepsilon})$ for class (S1) and a weakened form for arbitrary index sequences. Its first part is the theorem of Lan and Zhou, quoted with the displays of arXiv:1507.02000v3, whose sum convention gives the same function and bounds as the journal version [25]. On their instance (A1.7), m components act on disjoint blocks of dimension $\tilde{n}$, each a Nesterov chain f_i, convex with an L_i-Lipschitz gradient, $L_i = \mu(\mathcal{Q} - 1)$, plus its own regularizer; $n = m\tilde{n}$ is the total dimension, and each block has condition number at most $\mathcal{Q}$.

Theorem A1.8 (Lan–Zhou lower bound; Theorem 3 and Corollary 3 of [25])**.** *Let $\mathcal{Q}$ be the parameter of the instance (A1.7) and $q = \frac{\sqrt{\mathcal{Q}}-1}{\sqrt{\mathcal{Q}}+1}$. The iterates of any randomized incremental gradient method of class (S1) (Definition A1.2) applied to (A1.7) satisfy*

$$\frac{\mathbb{E}\left\|x^k - x^*\right\|^2}{\left\|x^0 - x^*\right\|^2} \;\geq\; \frac{1}{2}\exp\Big(-\frac{4k\sqrt{\mathcal{Q}}}{m(\sqrt{\mathcal{Q}}+1)^2 - 4\sqrt{\mathcal{Q}}}\Big) \qquad \text{for all} \;\; n \geq \underline{n}(m,k),$$

$$\underline{n}(m,k) := \frac{m \log\left[(1-(1-q^2)/m)^k/2\right]}{2\log q},$$

where $n = m\tilde{n}$ is the total dimension. Consequently the number of gradient evaluations needed by such a method to find $\bar{x}$ with $\mathbb{E}\left\|\bar{x} - x^*\right\|^2 \leq \varepsilon$ cannot be smaller than

$$\Omega\Big\{\big(m + \sqrt{m\,\mathcal{C}}\big)\log\frac{\left\|x^0 - x^*\right\|^2}{\varepsilon}\Big\}, \qquad \mathcal{C} = \frac{L}{\mu},\; L = \sum_{i=1}^m L_i,$$

if n is sufficiently large; precisely (their display (3.45) in arXiv:1507.02000v3) $k \geq \big[\frac{m}{2}\big(\frac{\sqrt{\mathcal{Q}}}{2}+1\big) - 1\big]\log\frac{\left\|x^0-x^*\right\|^2}{2\varepsilon}$.

[Appendix A1 of the full report, p. 148]

Written as an average of m components, m times the summands of (A1.7), the instance has $R^2 = m\mu\mathcal{Q}$, so that $n_{\text{book}} = m$, $\kappa = m\mathcal{Q}$ and, as $\mathcal{C} = m(\mathcal{Q}-1)$, $m + \sqrt{m\mathcal{C}} \asymp n_{\text{book}} + \sqrt{n_{\text{book}}\kappa}$: this is the bound $\Omega((n+\sqrt{n\kappa})\log\frac{1}{\varepsilon})$ for class (S1), the first part of A1, on instances with $\kappa \geq n$ and a dimension that grows with the number of queries.

The second part holds only in a restricted sense. Han, Xie and Zhang [17] prove a lower bound for a class $\mathcal{A}$ that, by their account, contains SAGA, L-SVRG and L-Katyusha: at each step, an algorithm of $\mathcal{A}$ samples one index i.i.d. from a fixed distribution, may query the proximal incremental first-order oracle (PIFO, which returns the value, the gradient and a proximal step of a component) of the sampled component at all previous iterates, computes the full gradient at the current point with an independent probability at most c_0/n, for a constant c_0 (the full gradient at the initial point is free), and takes its iterate in the span of the previous iterates and of the answers, projected on the domain. In their theorem, L is the smoothness constant of the components (our R^2), and R is not ours: $2R$ bounds the diameter of the domain.

Theorem A1.12 (Han–Xie–Zhang lower bound; Theorem E.6 of [17]). *Let $n \geq 2$ and consider the class $\mathcal{F}^{\mathrm{C}}(R, L, \mu)$ of finite sums $f = \frac{1}{n}\sum_i f_i$ on a closed convex domain $\mathcal{X}$ of diameter at most $2R$, with L-smooth components and μ-strongly convex f, $\kappa = L/\mu \geq 2$, and let $\varepsilon \leq LR^2/4$ (the hard instance is constrained to the ball $\{\|x\| \leq R\}$, which contains $x_0 = 0$ and x^*; its components are individually μ-strongly convex, their Proposition E.13). The number of PIFO queries that algorithms of class $\mathcal{A}$ (their Definition 3.1, which by their account contains SAGA, L-SVRG and L-Katyusha) need to find $\hat{x} \in \mathcal{X}$ with $\mathbb{E}f(\hat{x}) - \min_{\mathcal{X}} f < \varepsilon$ is*

$$\mathfrak{m}_*^{\mathrm{C}}(\varepsilon, R, L, \mu) = \begin{cases} \Omega\big((n+\sqrt{\kappa n})\log(1/\varepsilon)\big) & \text{for } \kappa = \Omega(n), \\ \Omega\big(n + \frac{n}{1+(\log(n/\kappa))_+}\log(1/\varepsilon)\big) & \text{for } \kappa = O(n), \end{cases}$$

and the construction only requires the dimension to be $O\big(1+\sqrt{\kappa/n}\,\log(1/\varepsilon)\big)$.

[Appendix A1 of the full report, p. 150]

For the methods of class $\mathcal{A}$, and, up to logarithmic factors, for the oblivious methods of part (c) of Theorem A1.11 (Hannah–Liu–O'Connor–Yin bounds; [18]), the lower bound is $\Omega\big(n + (\frac{n}{1+(\ln(n/\kappa))_+} + \sqrt{n\kappa})\ln\frac{1}{\varepsilon}\big)$: the $\sqrt{n\kappa}$ term is untouched, and full gradients can divide $n\log\frac{1}{\varepsilon}$ by at most $1 + (\ln(n/\kappa))_+$, a gain only when $\kappa \ll n$, never below the bare n; Prox-SVRG with the parameters of Theorem A1.11(b) attains this bound when $\kappa = O(n)$. Without these restrictions, only an additive n is proved for randomized algorithms, while deterministic algorithms, cyclic incremental methods included, can need $\Omega(n\sqrt{\kappa}\log\frac{\varepsilon_0}{\kappa\varepsilon})$ when $\kappa > 73$, by Theorem A1.9 (Woodworth–Srebro lower bounds; Theorems 4 and 8 of [29]), which excludes for them any gain over $n\log\frac{1}{\varepsilon}$ at small ε; whether randomly reshuffled span methods, or deterministic ones when $\kappa \leq 73$, can beat $n\log\frac{1}{\varepsilon}$ is not settled by these sources.

A hard instance with a closed-form minimizer. The instance of Theorem A1.8 (Lan–Zhou lower bound; Theorem 3 and Corollary 3 of [25]) gives each component its own regularizer, which forces $\kappa \geq n$. The instance (S) of Appendix A1 keeps one Nesterov chain per component but shares the regularizer among all components, as in [1], so that every $\kappa > 1$ is allowed. With n blocks $\theta^{(i)} \in \mathbb{R}^m$ and $A_m(\zeta)$ the $m \times m$ tridiagonal matrix with -1 off the diagonal and 2 on it, except $(A_m(\zeta))_{mm} = \zeta \in [1,3]$ (see (A1.2)),

$$\begin{aligned} &c = \frac{\mu(\kappa-1)}{4}, \qquad \phi(x) = \frac{1}{2}\langle A_m(\zeta)x, x\rangle - x_1 \quad (x \in \mathbb{R}^m), \\ &f_i(\theta) = c\,\phi(\theta^{(i)}) + \frac{\mu}{2}\|\theta\|^2, \quad i = 1, \dots, n. \end{aligned} \tag{A1.8}$$

The instance (H) of Section 2, used in Figures A3, A2 and A4, has instead a regularizer local to each block and $\kappa > n$. The constants and the minimizer of (S) are explicit.

Proposition A1.13 (constants of the instance). *Let $Q := 1 + \frac{\kappa-1}{n}$. For the instance (A1.8):*

(a) each f_i is μ-strongly convex and R^2-smooth with $R^2 = \mu\kappa$ (its exact Lipschitz constant is $\mu + c\,\lambda_{\max}(A_m(\zeta)) \le \mu\kappa$, with equality only in the limit $m \to \infty$);

(b) $\nabla^2 F = \operatorname{diag}\left(\frac{c}{n}A_m(\zeta) + \mu I, \dots, \frac{c}{n}A_m(\zeta) + \mu I\right)$ is block diagonal; F is μ-strongly convex, its smallest Hessian eigenvalue is $\mu_F := \mu + \frac{c}{n}\lambda_{\min}(A_m(\zeta)) = \mu\big(1 + \frac{Q-1}{4}\lambda_{\min}(A_m(\zeta))\big) \ge \mu$ and its smoothness constant is $L_F = \mu + \frac{c}{n}\lambda_{\max}(A_m(\zeta)) \le \mu Q$;

(c) hence $\kappa = R^2/\mu = 1 + n(Q-1)$ and

$$\frac{L_F}{\mu} \;\le\; Q = 1 + \frac{\kappa - 1}{n} \;\le\; \kappa, \tag{A1.10}$$

so that $(L_F/\mu)/\kappa \le \frac{1}{n} + \frac{1}{\kappa}$: the condition number of F is about n times smaller than κ as soon as $\kappa \gg n$.

[proof: Appendix A1 of the full report, p. 152]

On (S), full-gradient methods therefore depend on $L_F \le \mu Q$, about R^2/n when $\kappa \gg n$, rather than on R^2: gradient descent and AGD do not pay there the factors n and $\sqrt{n}$ of Table 1, which require aligned components, as in the aligned instance (A) of Figure A4b.

Proposition A1.14 (closed-form minimizer). *For the instance (A1.8), let $Q = 1 + \frac{\kappa-1}{n}$ and $q = \frac{\sqrt{Q}-1}{\sqrt{Q}+1}$.*

(a) (Geometric chain.) If

$$\zeta = \zeta_Q := \frac{\sqrt{Q}+3}{\sqrt{Q}+1} \in (1, 2), \tag{A1.11}$$

then the minimizer is $\eta_* = (x^*, \dots, x^*)$ with $x^*_j = q^j$, $j = 1, \dots, m$, so that

$$\|\eta_*\|^2 = n\,\frac{q^2(1-q^{2m})}{1-q^2}, \qquad F(\eta_*) = -\frac{c\,q}{2} = -\frac{\mu(\kappa-1)\,q}{8}.$$

(b) (Nesterov's chain, $\zeta = 2$.) The minimizer is $\eta_* = (x^*, \dots, x^*)$ with

$$x^*_j = \frac{q^j - q^{2(m+1)-j}}{1 - q^{2(m+1)}} = \frac{\sinh\big((m+1-j)\tau\big)}{\sinh\big((m+1)\tau\big)}, \qquad \tau = \log\frac{1}{q} = 2\,\operatorname{artanh}\frac{1}{\sqrt{Q}}, \quad j = 1, \dots, m.$$

(c) (Any $\zeta \in [1, 3]$.) x^* is the solution of the tridiagonal system $\big(A_m(\zeta) + \frac{4}{Q-1}I\big)x = e_1$, computable in $O(m)$ by the Thomas algorithm (forward elimination, backward substitution), which is numerically stable here because the matrix is symmetric positive definite and diagonally dominant.

In all cases $F(\theta) - F(\eta_*) = \frac{1}{2}\langle \nabla^2 F(\theta - \eta_*), \theta - \eta_*\rangle$ lies between $\frac{\mu_F}{2}\|\theta - \eta_*\|^2$ and $\frac{L_F}{2}\|\theta - \eta_*\|^2$, with μ_F and L_F as in Proposition A1.13(b).

[proof: Appendix A1 of the full report, p. 152]

Part (a) gives η_* in closed form, so that every error measured on (S) is exact; part (b), with $Q = 1 + 4c/\mu_b$, gives the minimizer of (H) (Section 2).

Started from $\theta_0 = 0$, a span method reveals at most one new coordinate of block j per query to component j by Lemma A1.16 (support under the span protocol), and the coordinates not yet revealed carry an explicit share of $\|\eta_*\|^2$ by Proposition A1.17 (tail bound). Averaging over the blocks by Jensen's inequality gives a bound for every index sequence, the $\sqrt{n\kappa}$ term of A1:

Proposition A1.18 (the $\sqrt{n\kappa}$ term for any index sequence). *On the instance (A1.8) with $\zeta = \zeta_Q$, and Q, q as in Proposition A1.14, consider any method of class (S2) (Definition A1.2) started at*

$\theta_0 = 0$. *After G gradient evaluations in total (full gradients counted as n), of which τ_j are queries to component j, every iterate θ in the span of the answers satisfies*

$$\frac{1}{n}\sum_{j=1}^{n} q^{2\tau_j} \;\ge\; q^{2G/n}, \qquad \textit{hence} \qquad \frac{\|\theta-\eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{q^{2G/n}-q^{2m}}{1-q^{2m}}. \tag{A1.13}$$

Consequently, if $q^{2m} \le \varepsilon < \frac{1}{2}$, reaching $\|\theta-\eta_*\|^2 \le \varepsilon\,\|\eta_*\|^2$ requires

$$G \;\ge\; \frac{n\log\frac{1}{2\varepsilon}}{2\log(1/q)} \;\ge\; \frac{n(\sqrt{Q}-1)}{4}\log\frac{1}{2\varepsilon} \;\ge\; \frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon} \quad (\textit{the last step for } \kappa-1\ge 2n). \tag{A1.14}$$

[proof: Appendix A1 of the full report, p. 154]

For the methods with coin flips, such as L-SVRG and L-Katyusha, the bound holds for the expected number of gradient evaluations (Appendix A1, after the proof). AGD on F, which needs $O(n\sqrt{Q}\log\frac{1}{\varepsilon})$ gradients since $L_F \le \mu Q$, attains its order when $\kappa \gtrsim n$. The sequence-free argument cannot produce the $n\log\frac{1}{\varepsilon}$ term, because a full gradient advances all n chains at once. With i.i.d. indices, the number of queries to each block is binomial, and the term appears:

Theorem A1.19 (Lan–Zhou bound on the instance)**.** *On the instance (A1.8) with $\zeta = \zeta_Q$, and Q, q as in Proposition A1.14, let $\theta_0 = 0$ and let the method belong to class (S1) (Definition A1.2) with a fixed sampling distribution p (uniform or not). Then for every $t \ge 0$,*

$$\frac{\mathbb{E}\,\|\theta_t-\eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{\left(1-\frac{1-q^2}{n}\right)^t - q^{2m}}{1-q^{2m}}, \tag{A1.15}$$

and, whenever $q^{2m} \le \frac{1}{2}\left(1-\frac{1-q^2}{n}\right)^t$,

$$\frac{\mathbb{E}\,\|\theta_t-\eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{1}{2}\Big(1-\frac{1-q^2}{n}\Big)^t \;\ge\; \frac{1}{2}\exp\Big(-\frac{4\sqrt{Q}\,t}{n(\sqrt{Q}+1)^2-4\sqrt{Q}}\Big). \tag{A1.16}$$

[proof: Appendix A1 of the full report, p. 155]

Corollary A1.20 (iteration count and dimension requirement)**.** *In the setting of Theorem A1.19, let $\varepsilon \in (0, \frac{1}{2})$ and assume the block length m satisfies*

$$m \;\ge\; \frac{\log(1/\varepsilon)}{2\log(1/q)}, \qquad \textit{for which it suffices that} \qquad m \;\ge\; \frac{\sqrt{Q}}{4}\log\frac{1}{\varepsilon} = \frac{1}{4}\sqrt{1+\frac{\kappa-1}{n}}\,\log\frac{1}{\varepsilon}. \tag{A1.17}$$

Then any method of class (S1) needs

$$t \;\ge\; \Big(\frac{n(\sqrt{Q}+1)^2}{4\sqrt{Q}}-1\Big)\log\frac{1}{2\varepsilon} \;\ge\; \Big(\frac{n+\sqrt{n(\kappa-1)}}{8}+\frac{n}{2}-1\Big)\log\frac{1}{2\varepsilon} = \Omega\Big((n+\sqrt{n\kappa})\log\frac{1}{\varepsilon}\Big) \tag{A1.18}$$

gradient evaluations to reach $\mathbb{E}\,\|\theta_t-\eta_*\|^2 \le \varepsilon\,\|\eta_*\|^2$. In particular $t \ge (n-1)\log\frac{1}{2\varepsilon}$ for every $\kappa > 1$, because $(\sqrt{Q}+1)^2 \ge 4\sqrt{Q}$.

[proof: Appendix A1 of the full report, p. 155]

This proves the first part of A1 on (S), for every $\kappa > 1$, under an explicit dimension condition: by (A1.17) each block must be longer than the number of touches that an accelerated method needs, of order $\frac{1}{4}\sqrt{\kappa/n}\log\frac{1}{\varepsilon}$ for $\kappa \gg n$, so that the total dimension nm is of the order of the number of queries certified. L-Katyusha is not in (S1), because of its full gradients: its lower bound on (S) is Proposition A1.18 (the $\sqrt{n\kappa}$ term for any index sequence), without the $n\log\frac{1}{\varepsilon}$ term.

SAGA and L-Katyusha on the instance (hypothesis A4). Hypothesis A4 predicts that the gradient counts of SAGA and L-Katyusha grow like κ and $\kappa^{1/2}$ when $\kappa \gg n$. The chain bounds above are at most of order $(n+\sqrt{n\kappa})\log\frac{1}{\varepsilon}$; for SAGA, the exponent 1 comes instead from its mean iterate, which moves like gradient descent whatever the memory:

Proposition A1.22 (mean iterate of SAGA on quadratics). *Let the f_i be convex quadratics with Hessians $H_i \succeq 0$, let $\bar{H} = \frac{1}{n}\sum_i H_i = \nabla^2 F$ have eigenpairs $(\nu_l, w_l)_{l=1}^d$ (w_l orthonormal) and $\mu_F = \lambda_{\min}(\bar{H})$, and for $\nu' \geq \mu_F$ let $P_{\nu'} = \sum_{l:\nu_l \leq \nu'} w_l w_l^\top$ be the orthogonal projector onto the sum of the eigenspaces of $\bar{H}$ with eigenvalues at most ν'. For SAGA with step $\gamma > 0$, uniform sampling and any initial memory $z^{(0)}$ (deterministic, or random and independent of the sampling), $e_t = \theta_t - \eta_*$ satisfies $\mathbb{E}[e_t] = (I-\gamma\bar{H})^t e_0$; hence, for every $\nu' \in [\mu_F, 1/\gamma]$,*

$$\mathbb{E}\,\|\theta_t - \eta_*\|^2 \;\geq\; \big\|(I-\gamma\bar{H})^t(\theta_0-\eta_*)\big\|^2 = \sum_{l=1}^{d}(1-\gamma\nu_l)^{2t}\,\langle \theta_0-\eta_*, w_l\rangle^2 \;\geq\; (1-\gamma\nu')^{2t}\,\|P_{\nu'}(\theta_0-\eta_*)\|^2\,. \tag{A1.20}$$

With $\nu' = \mu_F$ and a simple smallest eigenvalue with unit eigenvector $v_{\min}$, the last term is $(1-\gamma\mu_F)^{2t}\,\langle\theta_0-\eta_*, v_{\min}\rangle^2$. [proof: Appendix A1 of the full report, p. 158]

For n identical components this is the argument of Proposition B3.17 (lower bound for SAGA with any constant step); here it holds on every quadratic instance. Combined with the proof of Proposition 5.9, run with $\mu_F = \lambda_{\min}(\nabla^2 F)$ in place of μ, it gives a two-sided bound:

Corollary A1.23 (SAGA sandwich). *Consider an instance with convex quadratic components such that $\lambda_{\max}(H_i) \leq R^2$ for all i (each f_i is then convex and R^2-smooth), with μ_F and the projectors $P_{\nu'}$ of Proposition A1.22. Let $\kappa_{\text{eff}} = R^2/\mu_F$, $\gamma = 1/(4R^2)$, and let $T_{\text{SAGA}}(\varepsilon)$ be the number of gradient evaluations (the n of an initial sweep $z_i^{(0)} = f_i'(\theta_0)$ included) that SAGA with step γ and uniform sampling needs to reach $\mathbb{E}\,\|\theta_t-\eta_*\|^2 \leq \varepsilon\,\|\theta_0-\eta_*\|^2$.*

(a) (Lower bound, any initial memory.) If $\|P_{\nu'}(\theta_0-\eta_*)\|^2 \geq \rho\,\|\theta_0-\eta_*\|^2$ for some $\nu' \in [\mu_F, 4R^2]$ and $\rho \in (0,1]$, then for $\varepsilon < \rho$

$$T_{\text{SAGA}}(\varepsilon) \;\geq\; \Big(\frac{2R^2}{\nu'} - \frac{1}{2}\Big)\log\frac{\rho}{\varepsilon},$$

i.e. $(\kappa - \frac{1}{2})\log\frac{\rho}{\varepsilon}$ for $\nu' = 2\mu$ and $(2\kappa_{\text{eff}} - \frac{1}{2})\log\frac{\rho}{\varepsilon}$ for $\nu' = \mu_F$.

(b) (Upper bound.) With the initialization $z_i^{(0)} = f_i'(\theta_0)$ of Proposition 5.9 of [9], the number of iterations needed is at most

$$\max\Big\{3n, \frac{16\kappa_{\text{eff}}}{3}\Big\}\log\frac{1+n/4}{\varepsilon} \qquad (\text{add } n \text{ gradient evaluations for the initial sweep}),$$

and with $z^{(0)} = 0$ the same bound holds with $1+n/4$ replaced by $1 + \frac{3\sum_i \|f_i'(\eta_*)\|^2}{16R^4\|\theta_0-\eta_*\|^2}$.

[proof: Appendix A1 of the full report, p. 159]

On (S), the eigenvectors of the chain are sampled sine waves, and those of curvature at most 2μ carry a fixed share of the initial error as long as the chain is long compared with the decay length $\approx \sqrt{Q}/2$ of η_*:

Lemma A1.24 (slow-mode mass of the initial error). *Let $m \geq 2$, $\zeta \in [1,2]$ and $s \in (0,1]$. Let $0 < \lambda_1 < \cdots < \lambda_m < 4$ be the eigenvalues of the chain matrix $A_m(\zeta)$ of (A1.2) with orthonormal*

eigenvectors $v_1, \dots, v_m$, and let $\Pi_s = \sum_{k:\lambda_k \le s} v_k v_k^\top$ be the orthogonal projector onto the eigenspaces with eigenvalue at most s. For $x^* = (A_m(\zeta) + sI)^{-1} e_1$ and $K_0 := \lfloor (m+1)\sqrt{s}/\pi \rfloor$,

$$\|\Pi_s x^*\|^2 \;\ge\; \frac{7\pi^2(1-s/4)}{96\, m\, s^2 (2m+1)^2}\, K_0\big(4K_0^2 - 1\big). \tag{A1.23}$$

If moreover $\zeta = \zeta_Q$ and $s = \frac{4}{Q-1}$ (so that $x^* = (q^j)_{j \le m}$ is the block of η_*, Proposition A1.14(a)), the slow-mode fraction $\rho := \|\Pi_s x^*\|^2 / \|x^*\|^2$ satisfies, with $y := (m+1)\sqrt{s}/\pi$,

$$\rho \;\ge\; \frac{7(1-s/4)}{192\pi} \cdot \frac{K_0(4K_0^2-1)}{y^3} \;\ge\; \begin{cases} 1/105 & \text{if } y \ge 2, \\ 1/310 & \text{if } y \ge 1. \end{cases} \tag{A1.24}$$

On the instance (A1.8) with $\theta_0 = 0$, $\|P_{2\mu}(\theta_0 - \eta_*)\|^2 = \rho\, \|\theta_0 - \eta_*\|^2$, where $P_{2\mu}$ is the projector of Proposition A1.22 for $\nu' = 2\mu$. [proof: Appendix A1 of the full report, p. 161]

These results and the bound of Theorem 11 of [24] for L-Katyusha (Section 4.5 recalls its iteration, its mirror sequence and its reference point, written z_t and w_t there and z^k and w^k below, and its parameters θ_1 and σ), converted into the distance of its mirror sequence in (A1.21)–(A1.22) with an explicit constant $C_0 \ge 1$ that depends only on n, κ and L_F/μ and enters the counts only through $\log \frac{C_0}{\varepsilon}$, give the prediction tested in Section 5.9:

Proposition A1.26 (testable prediction of hypothesis A4). *Fix $n \ge 2$, $m \ge 2$, μ and a relative accuracy $\varepsilon \in (0, \frac{1}{105})$, start at $\theta_0 = 0$, and let κ vary. On the instance (A1.8) with $\zeta = \zeta_Q$ (Q, q as in Proposition A1.14; μ_F, L_F as in Proposition A1.13), let $T_{\mathrm{alg}}(\varepsilon)$ be the number of gradient evaluations (its expectation for L-Katyusha) that an algorithm needs to reach $\mathbb{E}\,\|\theta - \eta_*\|^2 \le \varepsilon\, \|\theta_0 - \eta_*\|^2$: SAGA with $\gamma = \frac{1}{4R^2}$ and $z_i^{(0)} = f_i'(0)$, the n gradients of this sweep being counted; L-Katyusha with the parameters of Theorem 11 of [24], θ being its mirror sequence z^k, charged n gradients for $F'(w^0)$ and, per iteration, one gradient plus n when the reference point is refreshed (probability $p = \frac{1}{n}$; the gradients $f_i'(w^k)$ of each full pass are stored), i.e. 2 per iteration in expectation; GD and AGD on F, n per iteration. Let C_0 be the constant of (A1.21).*

(i) (Plateau, $\kappa \le n$.) For SAGA, under (A1.17) and $\varepsilon < q^2/2$,

$$n + (n-1)\log\frac{q^2}{2\varepsilon} \;\le\; T_{\mathrm{SAGA}}(\varepsilon) \;\le\; n + \max\Big\{3n, \frac{16\kappa}{3}\Big\} \log\frac{1+n/4}{\varepsilon} \;\le\; n + \frac{16n}{3}\log\frac{1+n/4}{\varepsilon},$$

both sides being $\Theta(n\log(1/\varepsilon))$ once $\varepsilon \ll q^2$ and $\varepsilon \ll 1/n$ (the upper bound is $\Theta(n\log(n/\varepsilon))$ in general): the plateau with slope ≈ 0 is a theorem for SAGA. For L-Katyusha only

$$\frac{n\log(1/(2\varepsilon))}{2\log(1/q)} \;\le\; T_{\mathrm{LK}}(\varepsilon) \;\le\; n + 16n\log\frac{C_0}{\varepsilon}, \qquad C_0 \le n + \tfrac{7}{3},$$

is proved (the lower bound under (A1.17)); the two sides are of the same order $n\log(1/\varepsilon)$ only when $\log(1/q) = O(1)$, i.e. when κ is of order n (for $\kappa = n$, $\log(1/q) = 2\,\mathrm{artanh}\,\big((2-\frac{1}{n})^{-1/2}\big)$, which decreases to $\log(3+2\sqrt{2}) \approx 1.763$ as $n \to \infty$: 1.81 for $n = 16$, 1.77 for $n = 100$). For $\kappa \ll n$, $\log(1/q) \approx \log\frac{4n}{\kappa-1}$, and a gain of this logarithmic order over $n\log(1/\varepsilon)$ is not excluded for L-Katyusha; it is realized by GD on F with step $1/L_F$, which contracts the error by $1 - \frac{\mu_F}{L_F} \le 1 - \frac{1}{Q} = \frac{\kappa-1}{n+\kappa-1} \approx \frac{\kappa-1}{n}$ per full gradient, in agreement with Theorem A1.11(c) and with item (ii) of the assessment of hypothesis (A1) in Section A1.2. In the upper bounds, the second term takes over at $\kappa = \frac{9}{16}n$ for SAGA ($\frac{16\kappa}{3} = 3n$) and at $\kappa = \frac{8}{3}n$ for L-Katyusha (its momentum parameter θ_1 reaches $\frac{1}{2}$).

(ii) (Power laws.) Let

$$\kappa_{\max}(m,\varepsilon) := 1 + n\min\Big\{\frac{16m^2}{\log^2(1/\varepsilon)} - 1,\ \frac{(m+1)^2}{\pi^2}\Big\}, \tag{A1.27}$$

where the first term is the sufficient form $m \geq \frac{\sqrt{Q}}{4}\log\frac{1}{\varepsilon}$ of the dimension condition (A1.17) solved for κ (so that $q^{2m} \leq \varepsilon$), and the second is the condition $y \geq 2$ of Lemma A1.24 solved for κ. In the window $4n+1 \leq \kappa \leq \kappa_{\max}(m,\varepsilon)$, with $\rho = \rho(m,\kappa)$ the slow-mode fraction of Lemma A1.24,

$$\Big(\kappa - \frac{1}{2}\Big)\log\frac{\rho}{\varepsilon} \ \leq\ T_{\mathrm{SAGA}}(\varepsilon) \ \leq\ n + \frac{16\kappa}{3}\log\frac{1+n/4}{\varepsilon}, \qquad \rho = \rho(m,\kappa) \geq \frac{1}{105}, \tag{A1.28}$$

$$\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon} \ \leq\ T_{\mathrm{LK}}(\varepsilon) \ \leq\ n + 4\sqrt{6}\,\sqrt{n\kappa}\,\log\frac{C_0}{\varepsilon}, \qquad C_0 \leq 1 + \frac{5}{8}Q, \tag{A1.29}$$

so that $T_{\mathrm{SAGA}} = \Theta(\kappa\log(1/\varepsilon))$ and $T_{\mathrm{LK}} = \Theta(\sqrt{n\kappa}\log(1/\varepsilon))$ up to the displayed constants and the $\log(1+\frac{5}{8}Q)$ of (A1.29): κ enters T_{SAGA} and T_{LK} with the exponents 1 and $\frac{1}{2}$ as orders of growth (not as the slope of a line fitted over the window, see Remark A1.27), and $T_{\mathrm{SAGA}}/T_{\mathrm{LK}} \asymp \sqrt{\kappa/n}$. GD on F (step $1/L_F$) and AGD on F have the same two orders of growth: $n(\frac{L_F}{4\mu} - \frac{1}{2})\log\frac{\rho}{\varepsilon} \leq T_{\mathrm{GD}} \leq \frac{nQ}{2}\log\frac{1}{\varepsilon}$ with $L_F \geq \mu(1+\frac{13}{20}(Q-1))$, and $\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon} \leq T_{\mathrm{AGD}} = O(n\sqrt{Q}\log\frac{1}{\varepsilon})$.

(iii) (Saturation.) For $\kappa \gg nm^2$ the chain is exhausted: $\mu_F/\mu = 1 + \frac{\kappa-1}{4n}\lambda_{\min}(A_m(\zeta_Q))$ grows linearly in κ, so

$$\kappa_{\mathrm{eff}} = \frac{R^2}{\mu_F} = \frac{\kappa}{1+\frac{\kappa-1}{4n}\lambda_{\min}(A_m(\zeta_Q))} \ \xrightarrow[\kappa\to\infty]{}\ \frac{4n}{\lambda_{\min}(A_m(1))} = \frac{n}{\sin^2\frac{\pi}{2(2m+1)}} \approx \frac{4n(2m+1)^2}{\pi^2}, \tag{A1.30}$$

because $\zeta_Q \to 1$ and $\lambda_{\min}$ is continuous in the matrix; for every finite κ, Lemma A1.3 gives the two-sided bound $\frac{\kappa}{1+\frac{\kappa-1}{n}\sin^2\frac{\pi}{2(m+1)}} \leq \kappa_{\mathrm{eff}} \leq \frac{\kappa}{1+\frac{\kappa-1}{n}\sin^2\frac{\pi}{2(2m+1)}}$. Then $(2\kappa_{\mathrm{eff}} - \frac{1}{2})\log\frac{\delta}{\varepsilon} \leq T_{\mathrm{SAGA}}(\varepsilon) \leq n + \frac{16\kappa_{\mathrm{eff}}}{3}\log\frac{1+n/4}{\varepsilon}$ for $\varepsilon < \delta$, with the single-mode overlap $\delta = \langle x^*, v_1\rangle^2 / \|x^*\|^2$ (x^*, v_1 as in Lemma A1.24), which tends to $\delta_\infty(m) > 0$ (Remark A1.25): T_{SAGA} saturates at $\Theta(\kappa_{\mathrm{eff}}\log(1/\varepsilon)) = \Theta(nm^2\log(1/\varepsilon))$, while T_{LK} is no longer bounded below by $\sqrt{n\kappa}$ (the hypothesis $q^{2m} \leq \varepsilon$ of Proposition A1.18 fails) and its behaviour depends on the strong-convexity constant used to set the parameter σ of L-Katyusha. Since F is μ_F-strongly convex, Theorem 11 of [24] also applies with $\sigma = \mu_F/R^2$ (the tuning used in the experiments, where $\mu = \lambda_{\min}(\nabla^2 F)$ is passed to the method): (A1.21)–(A1.22) then hold with μ_F and κ_{eff} in place of μ and κ, so that $T_{\mathrm{LK}} = O\big((n+\sqrt{n\kappa_{\mathrm{eff}}})\log\frac{C_0}{\varepsilon}\big)$ with C_0 bounded in κ, and T_{LK} saturates at $O(nm\log\frac{C_0}{\varepsilon})$. With the nominal $\sigma = \mu/R^2$ of (A1.22) no saturation is proved: the upper bound keeps growing like $\sqrt{n\kappa}$, and the momentum $\theta_1 = \sqrt{2n\mu/(3R^2)}$ tends to 0, whereas the value matched to the curvature, $\sqrt{2n\mu_F/(3R^2)}$, does not. Thus T_{SAGA}, and T_{LK} with $\sigma = \mu_F/R^2$, stay bounded as $\kappa \to \infty$, and their log-log slopes against κ tend to 0.

[proof: Appendix A1 of the full report, p. 163]

Part (ii) certifies the exponents 1 and $\frac{1}{2}$ as orders of growth, not as slopes. For the instance of Figure A4b ($n = 16$, $m = 256$, $\varepsilon = 10^{-8}$) the window is $65 \leq \kappa \leq \kappa_{\max} \approx 4.9\cdot 10^4$, i.e. 2.9 decades; with ρ and C_0 replaced by their bounds $\frac{1}{105}$ and $1+\frac{5}{8}Q$, the two sides of (A1.28) differ there by a factor of about 7.8, and those of (A1.29) by 88 to 115, so that they constrain the end-to-end slope only to $[0.69, 1.31]$ and $[-0.17, 1.22]$, as Remark A1.27 (what to expect in a plot) details; the slopes

Method or class	Component gradients to reach accuracy ε	Validity and source
GD, step $1/R^2$	$n\kappa \log \frac{1}{\varepsilon}$ $(n\frac{L_F}{\mu} \log \frac{1}{\varepsilon}$ with step $1/L_F)$	all n, κ; [9, Section 5.4.4]
AGD	$n\sqrt{\kappa} \log \frac{1}{\varepsilon}$ $(n\sqrt{L_F/\mu} \log \frac{1}{\varepsilon}$ with $L_F)$	all n, κ; [9, Section 5.2.5]
SAGA, $\gamma = \frac{1}{4R^2}$	$n + \max\{3n, \frac{16\kappa}{3}\} \log \frac{1+n/4}{\varepsilon}$	all n, κ; Proposition 5.9; Corollary A1.23
L-SVRG, $\eta = \frac{1}{6R^2}$, $p = \frac{1}{n}$	$n + 3\max\{6\kappa, 2n\} \log \frac{\Phi_0}{\varepsilon}$	all n, κ; [24, Theorem 5]
L-Katyusha	$n + (12n + 15\sqrt{n\kappa}) \log \frac{3R^2(n+3)\|\theta_0 - \eta_*\|^2}{8\varepsilon}$	all n, κ and $\varepsilon \le \frac{3R^2(n+3)}{16} \|\theta_0 - \eta_*\|^2$; Corollary A3.12
class (S1)	$\Omega\big((n + \sqrt{n\kappa}) \log \frac{1}{\varepsilon}\big)$	dimension $\gtrsim$ number of queries; Theorem A1.8; Corollary A1.20
class (S2), instance (S)	$\frac{\sqrt{n(\kappa-1)}}{8} \log \frac{1}{2\varepsilon}$	$\kappa - 1 \ge 2n$, $q^{2m} \le \varepsilon < \frac{1}{2}$; Proposition A1.18
any randomized algorithm	$\Omega\big(n + \sqrt{n\kappa} \log(\frac{\varepsilon_0}{\varepsilon}\sqrt{n/\kappa})\big)$	$\kappa > 161n$, no span assumption; Theorem A1.9

Table 1: Guarantees for every instance of the class of Proposition 5.9 (top) and lower bounds (bottom), in component gradients. The accuracy criteria differ only inside the logarithm, by factors polynomial in n and κ (for instance $\frac{3\kappa(n+3)}{4}$ when the L-Katyusha criterion is converted into the relative squared distance by strong convexity): relative squared distance (SAGA, instance (S)), the Lyapunov function Φ of [24, Theorem 5] (L-SVRG), $\mathbb{E}[F(y_t)] - F(\eta_*)$ (L-Katyusha, y_t its gradient sequence, Section 4.5), initial gap ε_0 (last row). Counting convention, that of the experiments (Section 2): the SAGA, L-SVRG and L-Katyusha counts include the initial full gradient, and L-SVRG and L-Katyusha are charged 2 gradients per iteration ($f'_{i(t)}$ at the query point and at the reference point) and n per refresh of the reference point, i.e. 3 per iteration in expectation. With stored reference gradients, the convention of T_{LK} in Appendix A1, an iteration costs 2 in expectation, which gives $n + (8n + 10\sqrt{n\kappa}) \log(\cdot)$ for L-Katyusha and $n + 2\max\{6\kappa, 2n\} \log(\cdot)$ for L-SVRG. The SAGA and L-SVRG counts treat the number of iterations as a real number (rounding it up adds at most one iteration); the L-Katyusha count is that of an integer number of iterations, which its condition on ε ensures. For $\kappa \gg n$ the leading terms are $n\kappa$, $n\sqrt{\kappa}$, κ, κ and $\sqrt{n\kappa}$, the last one matching the lower bounds.

measured in Section 5.9 agree with these orders but are not implied by them. Figure A4b counts L-Katyusha as in Table 1 (3 gradients per iteration in expectation) and measures the distance of its gradient sequence y_t: this multiplies the upper bound of (A1.29) by at most $\frac{3}{2}$ and replaces C_0 by a constant $C_y = O(\kappa)$ (Appendix A1, after (A1.22)), without changing the exponents. Part (iii) proves that the counts saturate as $\kappa \to \infty$ only for SAGA and for L-Katyusha tuned with $\lambda_{\min}(\nabla^2 F)$, the tuning of the experiments, and proves nothing of the kind with the nominal μ; the slopes measured beyond $\kappa_{\max}$ are not predicted by part (ii). Logistic regression is not a hard instance: there is no chain, and its effective constants can be much better than R^2 and λ (Section 4.2).

Known and new. The lower bounds and the chain constructions [27, 1, 25] are known, and so is the geometric minimizer given by ζ_Q [25]. To our knowledge, the following are not in the sources we reviewed in this form: the restatement of the bounds with their exact oracle classes and the correction of hypothesis A1; self-contained proofs on a shared-regularizer instance valid for every $\kappa > 1$, including the bound for every index sequence (Proposition A1.18); the explicit dimension condition and window (Corollary A1.20, (A1.27)); and the SAGA sandwich (Corollary A1.23, with Lemma A1.24), which makes the exponent 1 a theorem on the instance (Proposition A1.26).

4.5 L-Katyusha: Lyapunov proof, negative momentum, and naive acceleration

Hypothesis A3 asks where the negative momentum of loopless Katyusha (L-Katyusha) [24] enters a Lyapunov proof written in the style of the book. For hypothesis A2, whether naive acceleration fails (Nesterov's momentum applied to the loopless SVRG estimator, without negative momentum), we ask why the same proof does not close in this case. Appendix A3 rewrites the proof of [24] in the notation of the book, under the following assumption.

Assumption A3.1 (convex R^2-smooth components, μ-strongly convex F)**.** *$F(\theta) = \frac{1}{n}\sum_{i=1}^n f_i(\theta)$ on $\mathbb{R}^d$, where each f_i is convex and R^2-smooth; we write $L := R^2$ so that L is the* per-sample *smoothness constant, as in Assumption 1 of [24]. F is μ-strongly convex, with minimizer η_* and optimal value $F_* := F(\eta_*)$. We write $\kappa := R^2/\mu = L/\mu \geq 1$ and $\sigma := \mu/L = 1/\kappa \in (0,1]$ (the quantity called σ in [24]).* [Appendix A3 of the full report, p. 169]

In this subsection $L = R^2$ is the per-sample smoothness constant, not the smoothness constant L_F of F, which Section 4.2 denotes L. L-Katyusha keeps a gradient sequence y_t, a mirror sequence z_t and a reference point w_t at which the full gradient $F'(w_t)$ is known. We keep the notation of [24] for its parameters: θ_1, θ_2 are momentum weights, not iterates, η is a step parameter, not the minimizer η_*, and z_t is the mirror sequence, not the SAGA memory. Let $\theta_1, \theta_2 > 0$ with $\theta_1 + \theta_2 \leq 1$, $p \in (0,1]$, and

$$\eta := \frac{\theta_2}{(1+\theta_2)\theta_1}. \tag{A3.2}$$

From $y_0 = z_0 = w_0 = \theta_0$, with $F'(w_0)$ computed, an iteration reads

$$x_t = \theta_1 z_t + \theta_2 w_t + (1-\theta_1-\theta_2) y_t, \tag{K1}$$
$$g_t = f'_{i(t)}(x_t) - f'_{i(t)}(w_t) + F'(w_t), \tag{K2}$$
$$z_{t+1} = \frac{1}{1+\eta\sigma}\Big(\eta\sigma\, x_t + z_t - \frac{\eta}{L}\, g_t\Big), \tag{K3}$$
$$y_{t+1} = x_t + \theta_1(z_{t+1} - z_t), \tag{K4}$$
$$w_{t+1} = y_t \text{ with probability } p, \qquad w_{t+1} = w_t \text{ otherwise}, \tag{K5}$$

where $i(t)$ is uniform on $\{1,\dots,n\}$ and the coin of (K5) is independent of $i(t)$ and of the past; $\mathbb{E}_{i(t)}$ denotes the expectation over both given (y_t, z_t, w_t), so that $\mathbb{E}_{i(t)}[g_t] = F'(x_t)$. Step (K3) is a mirror step, (K4) a gradient step, and (K1) couples them [3]. The term $\theta_2 w_t$ of (K1), the anchor, is the *negative momentum* of Katyusha [2]: it pulls the query point towards the point where the estimator has no variance. With $\theta_2 = 0$ and a free step η, $x_{t+1} = y_{t+1} + (1-\theta_1)(y_{t+1} - y_t)$ and the method is a variant of Nesterov's momentum applied to the loopless SVRG estimator, the naive acceleration of hypothesis A2; see Remark A3.16 (relation with Nesterov's method of the book). Theorem 11 of [24] takes

$$\theta_2 = \tfrac{1}{2}, \qquad \theta_1 = \min\Big\{\sqrt{\tfrac{2\sigma n}{3}}, \tfrac{1}{2}\Big\} = \min\Big\{\sqrt{\tfrac{2n}{3\kappa}}, \tfrac{1}{2}\Big\}, \qquad p = \tfrac{1}{n}, \qquad \eta = \frac{\theta_2}{(1+\theta_2)\theta_1} = \frac{1}{3\theta_1}, \tag{A3.3}$$

and its proof tracks the Lyapunov function

$$\Psi_t := Z_t + Y_t + W_t,$$
$$Z_t := \frac{L(1+\eta\sigma)}{2\eta}\,\|z_t - \eta_*\|^2, \qquad Y_t := \frac{1}{\theta_1}\big(F(y_t) - F_*\big), \qquad W_t := \frac{\theta_2(1+\theta_1)}{p\theta_1}\big(F(w_t) - F_*\big), \tag{A3.4}$$

the sum of a distance of the mirror sequence, of a function value of the gradient sequence and of a memory term, which measures the quality of the reference point. Finally, $D_F(w,x) := F(w) - F(x) - \langle F'(x), w - x\rangle \geq 0$ denotes the Bregman divergence of F.

The proof certifies a contraction of Ψ in conditional expectation at every step, for all admissible parameters, and a global rate for the parameters (A3.3).

Theorem A3.4 (contraction and rate of L-Katyusha; Lemma 10 and Theorem 11 of [24]). *Under Assumption A3.1, for any $\theta_1, \theta_2 > 0$ with $\theta_1 + \theta_2 \leq 1$, any $p \in (0,1]$, $\sigma = \mu/L$ and η as in (A3.2), the iterates of (K1)-(K5) and the potential (A3.4) satisfy, for all $t \geq 0$,*

$$\mathbb{E}_{i(t)}\big[\Psi_{t+1}\big] \;\leq\; \frac{1}{1+\eta\sigma}\, Z_t + \big(1-\theta_1(1-\theta_2)\big)\, Y_t + \Big(1-\frac{p\theta_1}{1+\theta_1}\Big)\, W_t \qquad \textit{(one-step contraction)}. \tag{A3.5}$$

With the parameters (A3.3), for all $t \geq 0$,

$$\mathbb{E}\big[\Psi_t\big] \leq (1-\rho)^t\, \Psi_0, \qquad \rho := \min\Big\{\frac{\sigma}{6\theta_1}, \frac{\theta_1}{2n}\Big\} = \begin{cases} \dfrac{1}{\sqrt{24\,n\kappa}} & \text{if } \kappa \geq \frac{8n}{3}, \\ \min\Big\{\dfrac{1}{3\kappa}, \dfrac{1}{4n}\Big\} \;\Big(\geq \dfrac{1}{8n}\Big) & \text{if } \kappa < \frac{8n}{3}, \end{cases} \tag{A3.6}$$

so that in all cases $\rho \geq \min\{1/(4n), 1/\sqrt{24n\kappa}\}$ (global rate).

[proof: Appendix A3 of the full report, p. 175]

We write c_Z, c_Y and c_W for the three coefficients of (A3.5), so that $\mathbb{E}_{i(t)}[\Psi_{t+1}]$ is at most $\max\{c_Z, c_Y, c_W\}\,\Psi_t$, a factor that the proof of (A3.6) bounds by $1-\rho$. The two terms of ρ are lower bounds on $1-c_Z$, the contraction rate of the distance Z_t of the mirror sequence, whose step $\eta/L = 1/(3\theta_1 L)$ grows as θ_1 decreases, and on $1-c_W$, that of the memory term W_t, refreshed with probability $\frac{1}{n}$; for $\kappa \geq \frac{8n}{3}$, the value $\theta_1 = \sqrt{2n/(3\kappa)}$ of (A3.3) balances them up to a factor 2. Along L-Katyusha runs, with $\mathbb{E}_{i(t)}[\Psi_{t+1}]$ computed exactly, (A3.5) holds at every recorded state, and the exact one-step contraction exponent is on average 2.8 to 4.7 times the certified exponent $-\log\max\{c_Z, c_Y, c_W\}$ and 3.7 to 9.3 times $-\log(1-\rho)$ (three problems, 3 seeds; Section 5.7). In gradient evaluations, the rate gives the following count.

Corollary A3.12 (gradient complexity of L-Katyusha). *With the parameters (A3.3), ρ as in (A3.6) and $y_0 = z_0 = w_0 = \theta_0$, for all $t \geq 0$,*

$$\begin{aligned} &\mathbb{E}\big[F(y_t) - F_*\big] \leq \theta_1(1-\rho)^t \Psi_0, \qquad \mathbb{E}\,\|z_t - \eta_*\|^2 \leq \frac{2\eta}{L(1+\eta\sigma)}(1-\rho)^t \Psi_0, \\ &\qquad\qquad \theta_1 \Psi_0 \leq \frac{3L(n+3)}{8}\,\|\theta_0 - \eta_*\|^2 . \end{aligned} \tag{A3.20}$$

Consequently $\mathbb{E}[F(y_t) - F_*] \leq \varepsilon$ as soon as $t \geq \big(4n + \sqrt{24n\kappa}\big)\log\frac{3L(n+3)\|\theta_0-\eta_*\|^2}{8\varepsilon}$. An iteration evaluates $f'_{i(t)}(x_t)$ and $f'_{i(t)}(w_t)$ in (K2) and, with probability $p = 1/n$, the n gradients of a full pass: $2 + pn = 3$ evaluations of some f'_i in expectation. If the n gradients $f'_i(w)$ computed during the pass are stored ($\mathcal{O}(nd)$ memory, as for the table of SAGA), $f'_{i(t)}(w_t)$ is a look-up and the count is $1 + pn = 2$ (this is the "$\mathcal{O}(1+pn)$ times per iteration" of [24]). Adding the n evaluations of $F'(w_0)$

at initialization, for $\varepsilon \le \frac{3L(n+3)}{16}\,\|\theta_0-\eta_*\|^2$ *the expected number of gradient evaluations is at most*

$$\begin{aligned} &n+\big(12\,n+3\sqrt{24}\,\sqrt{n\kappa}\big)\log\frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon}\\ &\qquad\le\ n+\big(12\,n+15\sqrt{n\kappa}\big)\log\frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon}\quad \textit{(without storage)},\\ &n+\big(8\,n+2\sqrt{24}\,\sqrt{n\kappa}\big)\log\frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon}\\ &\qquad\le\ n+\big(8\,n+10\sqrt{n\kappa}\big)\log\frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon}\quad \textit{(with storage)}, \end{aligned} \tag{A3.21}$$

and in both cases $\mathcal{O}\big((n+\sqrt{n\kappa})\log\frac{nL\|\theta_0-\eta_*\|^2}{\varepsilon}\big)$. [proof: Appendix A3 of the full report, p. 176]

The count without storage, three gradients per iteration in expectation, is the convention of the experiments and of Table 1; the count with storage is that of the bounds of Appendix A1. The condition on ε lets the constants absorb the rounding of the number of iterations to an integer. For $\kappa \gg n$, the leading term $\sqrt{n\kappa}\log\frac{1}{\varepsilon}$ is of the order of the lower bounds of Section 4.4.

The proof, and where the negative momentum acts. The proof of the one-step contraction (A3.5) chains five lemmas, L1–L5, which reproduce Lemmas 6–10 of [24] and use seven elementary inequalities, listed with their tightness ratios in Table 5 of Appendix A3. L1 (Lemma A3.6) bounds the variance of the estimator; L2 (Lemma A3.7) bounds the mirror step by the three-point identity; L3 (Lemma A3.8) bounds the gradient step by smoothness and Young's inequality; L4 (Lemma A3.10) is the identity (A3.14) for the refresh of w; and L5 (Lemma A3.11) combines them with strong convexity at x_t and convexity towards y_t. The form of L1 matters for hypothesis A3: it bounds the variance by a Bregman divergence, whose inner-product part the negative momentum cancels.

Lemma A3.6 (L1: unbiasedness and variance of the estimator; Lemma 6 of [24]). *For all* t, $\mathbb{E}_{i(t)}[g_t]=F'(x_t)$ *and*

$$\mathbb{E}_{i(t)}\,\big\|g_t-F'(x_t)\big\|^2=\frac{1}{n}\sum_{i=1}^n\big\|f_i'(x_t)-f_i'(w_t)\big\|^2-\big\|F'(x_t)-F'(w_t)\big\|^2\ \le\ \frac{1}{n}\sum_{i=1}^n\big\|f_i'(x_t)-f_i'(w_t)\big\|^2 \tag{A3.7}$$

$$\le\ 2L\,\big(F(w_t)-F(x_t)-\big\langle F'(x_t),w_t-x_t\big\rangle\big)\ =\ 2L\,D_F(w_t,x_t). \tag{A3.8}$$

We call (A3.7) *the* centering inequality *and* (A3.8) *the* co-coercivity bound*; their product is the* variance bound

$$\mathbb{E}_{i(t)}\,\big\|g_t-F'(x_t)\big\|^2\le 2L\,D_F(w_t,x_t). \tag{A3.9}$$

[proof: Appendix A3 of the full report, p. 172]

It is the one-sided (Bregman) form of co-coercivity in (A3.8), instead of the symmetric form used in the proof of Proposition 5.9, that produces this divergence; Section 5.7 measures the two steps of (A3.9) along trajectories. In the proof of L5 (Lemma A3.11), θ_2 is used for something other than bookkeeping at a single step, Step 6, the identity

$$\frac{\theta_2}{\theta_1}\big\langle F'(x_t),x_t-w_t\big\rangle-\frac{\theta_2}{\theta_1}\Big(F(w_t)-F(x_t)-\big\langle F'(x_t),w_t-x_t\big\rangle\Big)=-\frac{\theta_2}{\theta_1}\big(F(w_t)-F(x_t)\big), \tag{A3.19}$$

in which the first term comes from the anchor $\theta_2 w_t$ of (K1) and the second is the variance term, which enters through L3 and L1:

Proposition A3.14 (Bregman-gap cancellation by the negative momentum). *In the chain of Lemma A3.11, the variance of the estimator enters through L3 and L1 (Lemmas A3.8 and A3.6) as the term $-\frac{\theta_2}{\theta_1} D_F(w_t, x_t)$ with $D_F(w_t, x_t) = \big(F(w_t) - F(x_t)\big) + \langle F'(x_t), x_t - w_t\rangle$, and the weight θ_2 of w_t in (K1) enters, through the decomposition (A3.17), as the term $+\frac{\theta_2}{\theta_1}\langle F'(x_t), x_t - w_t\rangle$. The two inner products cancel identically, and the remainder $-\frac{\theta_2}{\theta_1}(F(w_t) - F(x_t))$ is absorbed as follows: $F(x_t)$ disappears because the coefficients of $F(x_t)$ sum to zero (Step 7 of the proof of Lemma A3.11), and $\frac{\theta_2}{\theta_1}(F(w_t) - F_*) = \frac{p}{1+\theta_1} W_t$ is paid by the reference-point potential, which contracts by (A3.14) because w is refreshed with probability p.* [proof: Appendix A3 of the full report, p. 177]

The inner-product part of $D_F(w_t, x_t)$ must be cancelled rather than absorbed. By smoothness, $D_F(w_t, x_t) \le \frac{L}{2}\|w_t - x_t\|^2$ is a squared distance at the scale L, while the certified decrease per step is of order $\rho\Psi_t$, with $\rho = 1/\sqrt{24n\kappa}$ when $\kappa \ge \frac{8n}{3}$: absorbing this term would require converting a squared distance into function values, at the price of a factor κ, which Proposition A3.19 (closing by smoothness and strong convexity costs κ) makes precise for the naive scheme; and the inner product $\langle F'(x_t), x_t - w_t\rangle$ has no sign and is not a difference of function values that a potential could pay. The negative momentum costs a constant factor: Y_t contracts by $\theta_1(1-\theta_2)$ instead of θ_1, half as fast for $\theta_2 = \frac{1}{2}$. It also ties the step to θ_2, as Remark A3.9 (the step size couples the two momenta) explains: for a general step, Young's inequality in L3 (Lemma A3.8) produces the variance coefficient $\nu/(2L\theta_1)$ with $\nu = \eta\theta_1/(1-\eta\theta_1)$, and the cancellation is exact only for $\nu = \theta_2$, that is, for the step (A3.2): a larger step leaves the residual $\frac{\nu-\theta_2}{\theta_1} D_F(w_t, x_t)$, the extra term of the naive scheme (Proposition A3.18) with $\nu - \theta_2$ in place of ν, and a smaller one wastes part of the pull. Along the L-Katyusha runs of Section 5.7, the cancelled term $\frac{\theta_2}{\theta_1}\langle F'(x_t), x_t - w_t\rangle$ is positive and larger than the certified one-step decrease $(1 - \max\{c_Z, c_Y, c_W\})\Psi_t$ on 9.9%, 5.2% and 19.3% of the recorded states of the three problems; only there could it not have been absorbed into that decrease, since where it is negative, $D_F(w_t, x_t) \le F(w_t) - F(x_t)$ is bounded by a difference of function values.

Naive acceleration (hypothesis A2). Without the anchor, the naive scheme is the same iteration with $\theta_2 = 0$ and a free step.

Definition A3.15 (naive accelerated SVRG). *The* naive scheme $\mathrm{N}(\theta_1, \eta, p)$ *is* (K1)–(K5) *with $\theta_2 = 0$ and the step $\eta > 0$ taken as a free parameter with $\eta\theta_1 \le 1$:*

$$
\begin{aligned}
&x_t = \theta_1 z_t + (1-\theta_1) y_t, \qquad g_t \text{ as in (K2)}, \qquad z_{t+1} = \frac{\eta\sigma x_t + z_t - \frac{\eta}{L} g_t}{1 + \eta\sigma}, \\
&y_{t+1} = x_t + \theta_1(z_{t+1} - z_t), \qquad w_{t+1} = \begin{cases} y_t & w.p.\ p, \\ w_t & w.p.\ 1-p. \end{cases}
\end{aligned}
\tag{A3.22}
$$

We set $\nu := \frac{\eta\theta_1}{1-\eta\theta_1} \in (0, \infty]$, with $\nu = \infty$ in the limiting case $\eta\theta_1 = 1$; this case is admissible only with exact gradients $g_t = F'(x_t)$ (Proposition A3.17), because the Young step of L3 (Lemma A3.8) requires $\nu < \infty$ (Remark A3.9). The potential is the same Ψ_t as in (A3.4) (with W_t defined with an arbitrary constant $a_W \ge 0$ in place of $\frac{\theta_2(1+\theta_1)}{p\theta_1}$, which is 0 when $\theta_2 = 0$).

[Appendix A3 of the full report, p. 178]

With exact gradients, the same lemmas prove acceleration, so that what fails with the estimator is the treatment of its variance.

Proposition A3.17 (acceleration with exact gradients and $\theta_2 = 0$). *Let $g_t = F'(x_t)$ in (A3.22) (full gradient), $\theta_2 = 0$, $\eta = 1/\theta_1$ (so $\eta\theta_1 = 1$ and $\nu = \infty$: the limiting case of Definition A3.15) and*

$\theta_1 = \sqrt{\sigma} = \sqrt{\mu/L}$. *Then for all* t,

$$Z_{t+1} + Y_{t+1} \le \frac{1}{1+\sqrt{\sigma}} Z_t + (1-\sqrt{\sigma})\, Y_t \le \Big(1 - \frac{1}{2}\sqrt{\frac{\mu}{L}}\Big)(Z_t + Y_t), \tag{A3.23}$$

hence $F(y_t) - F_* \le \sqrt{\mu/L}\,(1 - \frac{1}{2}\sqrt{\mu/L})^t (Z_0 + Y_0)$*: the accelerated rate, as for (5.16)–(5.17) in the book.* [proof: Appendix A3 of the full report, p. 178]

With the estimator, Steps 1–5 and 7 of the proof of L5 (Lemma A3.11) go through, but no step produces the inner product that cancelled the variance term.

Proposition A3.18 (one-step inequality of the naive scheme)**.** *For the scheme* $\mathrm{N}(\theta_1, \eta, p)$ *of Definition A3.15 with* $\eta\theta_1 < 1$ *(so that* $\nu < \infty$*), for all* t,

$$\begin{aligned} \mathbb{E}_{i(t)}\big[Z_{t+1} + Y_{t+1}\big] &\le \frac{1}{1+\eta\sigma} Z_t + (1-\theta_1)\, Y_t + \frac{\nu}{2L\theta_1} \mathbb{E}_{i(t)} \left\| g_t - F'(x_t) \right\|^2 \\ &\le \frac{1}{1+\eta\sigma} Z_t + (1-\theta_1)\, Y_t + \frac{\nu}{\theta_1} D_F(w_t, x_t), \end{aligned} \tag{A3.24}$$

where the last inequality is the variance bound (A3.9). The extra term $\frac{\nu}{\theta_1} D_F(w_t, x_t)$ *is nonnegative; under Assumption A3.1 it vanishes only when* $w_t = x_t$*, since* $D_F(w_t, x_t) \ge \frac{\mu}{2} \left\| w_t - x_t \right\|^2$ *by the* μ*-strong convexity of* F. [proof: Appendix A3 of the full report, p. 179]

The decrease of the potential that (A3.24) can certify per step is at most $\frac{\eta\sigma}{1+\eta\sigma} Z_t + \theta_1 Y_t$, plus, with a reference term $a_W(F(w_t) - F_*)$, the expected decrease $p a_W(F(w_t) - F_*) - p a_W(F(y_t) - F_*)$ of that term; none of these involves the query point x_t. Bounding the extra term through smoothness and strong convexity costs a factor κ:

Proposition A3.19 (closing by smoothness and strong convexity costs κ)**.** *Under Assumption A3.1, for the scheme* $\mathrm{N}(\theta_1, \eta, p)$ *of Definition A3.15 with* $\eta\theta_1 < 1$,

$$\frac{\nu}{\theta_1} D_F(w_t, x_t) \;\le\; 2\kappa\nu \Big[\frac{1}{\theta_1}\big(F(w_t) - F_*\big) + \frac{\eta}{1+\eta\sigma} Z_t + (1-\theta_1)\, Y_t \Big], \tag{A3.25}$$

so that, writing $W_t = a_W(F(w_t) - F_*)$ *with* $a_W > 0$ *and using (A3.14) with* $\theta_2(1+\theta_1)/(p\theta_1)$ *replaced by* a_W,

$$\mathbb{E}_{i(t)}[\Psi_{t+1}] \le \Big(\frac{1 + 2\kappa\nu\eta}{1+\eta\sigma}\Big) Z_t + \Big(1 - \theta_1 + 2\kappa\nu(1-\theta_1) + p a_W \theta_1\Big) Y_t + \Big(1 - p + \frac{2\kappa\nu}{\theta_1 a_W}\Big) W_t. \tag{A3.26}$$

The coefficient of Z_t *is* < 1 *iff* $2\kappa\nu\eta < \eta\sigma$*, i.e.* $\nu < \frac{1}{2\kappa^2}$*; the coefficient of* Y_t *is* < 1 *only if* $2\kappa\nu(1-\theta_1) < \theta_1$*, i.e.* $\nu < \frac{\theta_1}{2\kappa(1-\theta_1)}$*; the coefficient of* W_t *is* < 1 *iff* $a_W > \frac{2\kappa\nu}{p\theta_1}$ *(substituting this bound in the* Y*-coefficient sharpens the second condition to* $\nu < \frac{\theta_1}{2\kappa(2-\theta_1)}$*, which we do not need). No choice of* (θ_1, η, p, a_W) *makes (A3.26) certify a rate better than* $1 - \kappa^{-2}$ *per iteration: worse than gradient descent with step* $1/L$*, which contracts by* $1 - 1/\kappa$ *per iteration (per gradient evaluation, with* $p = 1/n$*, the comparison is* $1 - \frac{1}{3\kappa^2}$ *against* $1 - \frac{1}{n\kappa}$*, so the naive certificate is worse as soon as* $\kappa > n/3$*), let alone the accelerated* $1 - \Theta(1/\sqrt{n\kappa})$. [proof: Appendix A3 of the full report, p. 180]

This is a limit of the Lyapunov argument, not a lower bound for the naive scheme, as Remark A3.20 (other routes, and scope of the obstruction) details: bounding the extra term through Z_t instead of μ is no better, paying it by the reference term alone is impossible, and no lower bound specific to the scheme is known to us. It agrees with the literature: with one sample per iteration,

the guarantee of Nesterov's momentum on the SVRG estimator is that of SVRG [28], the anchor-free acceleration of [12] needs mini-batches of size of order $n^{2/3}$, and the semidefinite certificate of [22] is stated for negative-momentum weights of at least $\frac{1}{5}$. In the experiments at $\kappa = 10^4$ (one seed; Section 5.8), L-Katyusha with $\theta_2 = 0$ and otherwise unchanged parameters diverges on the instance (H) ($n = 16$ blocks of length 64) and converges on logistic regression, and Nesterov's momentum applied to the estimator, at its best grid setting, converges on both problems; where they converge, these anchor-free schemes need 8 to 20 times fewer gradient evaluations than L-SVRG. Re-tuned at every κ, Nesterov's momentum applied to the estimator has fitted slopes of about 0.8 in κ on block quadratics (Section 5.9). Prediction (P3) of Section A3.4, that a stable naive scheme keeps the exponent 1 of κ and does no better than L-SVRG, is thus refuted in its second part and not supported in its first: Proposition A3.18 and Proposition A3.19 describe a limitation of the certificate, not the observed behaviour.

Known and new. Theorem A3.4 and its constants are those of [24], the negative momentum is that of [2], and its role of cancelling an inner product of the variance bound is stated in [2, 31]. To our knowledge, the following are not in the sources we reviewed in this form: the rewriting of the proof as a chain of labelled inequalities with tightness ratios, the isolation of the cancellation as a single identity tied to the choice of the step (Proposition A3.14 and Remark A3.9), and Proposition A3.18 and Proposition A3.19, which quantify what the same argument certifies without the anchor.

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